

Rotor and Wake Aerodynamics

- Derivation of Vorticity Equation

Learning objectives

- To understand the derivation of the vorticity equation
 - Curl of momentum equation
- To understand the process of vorticity generation given a space distribution of forces
 - The actuator disc case in 2D and 3D

Start from Momentum Equation

- We start with the momentum equation (Navier-Stokes):

$$\frac{Du}{Dt} = \frac{\partial u}{\partial t} + u \cdot \nabla u = -\nabla p + f + \mu \nabla^2 u$$

- Using this identity, we take the curl of the momentum equation:

$$\nabla \times u = \omega$$

- We can derive the vorticity equation

$$\frac{D\omega}{Dt} = \nabla \times \frac{Du}{Dt} = \nabla \times \frac{\partial u}{\partial t} + \nabla \times (u \cdot \nabla u) = -\nabla \times \nabla p + \nabla \times f + \nabla \times (\mu \nabla^2 u)$$

1st term

$$\frac{D\omega}{Dt} = \nabla \times \frac{Du}{Dt} = \nabla \times \frac{\partial u}{\partial t} + \nabla \times (u \cdot \nabla u) = -\nabla \times \nabla p + \nabla \times f + \nabla \times (\mu \nabla^2 u)$$

1st term

$$\frac{D\omega}{Dt} = \nabla \times \frac{Du}{Dt} = \nabla \times \frac{\partial u}{\partial t} + \nabla \times (u \cdot \nabla u) = -\nabla \times \nabla p + \nabla \times f + \nabla \times (\mu \nabla^2 u)$$

- First we take a look at the 1st term of the equation.
- It can be simplified as:

$$\nabla \times \frac{\partial u}{\partial t} = \frac{\partial}{\partial t} (\nabla \times u) = \frac{\partial \omega}{\partial t}$$

3rd term

$$\frac{D\omega}{Dt} = \nabla \times \frac{Du}{Dt} = \nabla \times \frac{\partial u}{\partial t} + \nabla \times (u \cdot \nabla u) = -\nabla \times \nabla p + \nabla \times f + \nabla \times (\mu \nabla^2 u)$$

➤ The curl of a gradient = 0

$$-\nabla \times \nabla p = 0$$

5th term

$$\frac{D\omega}{Dt} = \nabla \times \frac{Du}{Dt} = \nabla \times \frac{\partial u}{\partial t} + \nabla \times (u \cdot \nabla u) = -\nabla \times \nabla p + \nabla \times f + \nabla \times (\mu \nabla^2 u)$$

➤ It can be simplified as:

$$\nabla \times (\mu \nabla^2 u) = \mu \nabla^2 \omega$$

2nd term

$$\frac{D\omega}{Dt} = \nabla \times \frac{Du}{Dt} = \nabla \times \frac{\partial u}{\partial t} + \nabla \times (u \cdot \nabla u) = -\nabla \times \nabla p + \nabla \times f + \nabla \times (\mu \nabla^2 u)$$

- This term is more complicated, first we rewrite the term between brackets using a vector identity:

$$(u \cdot \nabla u) = \frac{1}{2} \nabla(u \cdot u) - u \times (\nabla \times u)$$

2nd term

$$\frac{D\omega}{Dt} = \nabla \times \frac{Du}{Dt} = \nabla \times \frac{\partial u}{\partial t} + \nabla \times (u \cdot \nabla u) = -\nabla \times \nabla p + \nabla \times f + \nabla \times (\mu \nabla^2 u)$$

- Now we apply the curl:

$$\nabla \times \frac{1}{2} \nabla(u \cdot u) - \nabla \times (u \times (\nabla \times u))$$

- In the first term, again the curl of a gradient is taken (=0), the second term is simplified to:

$$-\nabla \times (u \times \omega)$$

2nd term

$$\frac{D\omega}{Dt} = \nabla \times \frac{Du}{Dt} = \nabla \times \frac{\partial u}{\partial t} + \nabla \times (u \cdot \nabla u) = -\nabla \times \nabla p + \nabla \times f + \nabla \times (\mu \nabla^2 u)$$

- This can be rewritten to (vector identity):

$$-\nabla \times (u \times \omega) = (u \cdot \nabla)\omega - (\omega \cdot \nabla)u + \omega(\nabla \cdot u) + u(\nabla \cdot \omega)$$

- We can apply these simplifications:

$$\omega(\nabla \cdot u) = 0 \text{ (incompressible flow assumption, divergence of velocity = 0)}$$

$$u(\nabla \cdot \omega) = 0 \text{ (divergence of curl = 0, } \nabla \cdot \nabla \times u = 0 \text{)}$$

Vorticity Equation

$$\frac{D\omega}{Dt} = \nabla \times \frac{Du}{Dt} = \frac{\partial \omega}{\partial t} + (u \cdot \nabla)\omega = \nabla \times f + (\omega \cdot \nabla)u + (\mu \nabla^2 \omega)$$

source/sinkvorticity turning
and stretchingdiffusion

- The material derivative of vorticity is composed of three terms:
1. A source/sink term, that is generated by a non-conservative force field ($\nabla \times f \neq 0$)
 2. A vorticity turning/stretching term $(\omega \cdot \nabla)u$, responsible for conserving the angular momentum.
 3. A vorticity diffusion term $\mu \nabla^2 \omega$.

Vorticity Equation

- In 2D, incompressible flow $(\omega \cdot \nabla)u = 0$, the vorticity equation is:

$$\frac{D\omega}{Dt} = \underbrace{\nabla \times f}_{\text{Source/sink}} + \underbrace{\mu \nabla^2 \omega}_{\text{diffusion}}$$

- This means that vorticity can only be introduced by a change in force field.

Creation of Vorticity in a rotor

- For an inviscid flow the vorticity equation simplifies to

$$\frac{D\omega}{Dt} = \nabla \times \mathbf{f}$$

- Vorticity can only be generated by a curl in the force field $\mathbf{f}$.

$$\mathbf{f} = [f_x \ f_y \ f_z]$$

Creation of Vorticity in a rotor

$$f = [f_x \ f_y \ f_z]$$

$$\nabla \times f = \begin{bmatrix} \frac{df_z}{dy} - \frac{df_y}{dz} \\ -\frac{df_z}{dx} + \frac{df_x}{dz} \\ \frac{df_y}{dx} - \frac{df_x}{dy} \end{bmatrix}$$

- Distribution of f in a 2D actuator disk with uniform loading



Creation of Vorticity in a rotor

- Force field $f = [f_x \ f_y \ f_z] = [f_x \ 0 \ 0]$

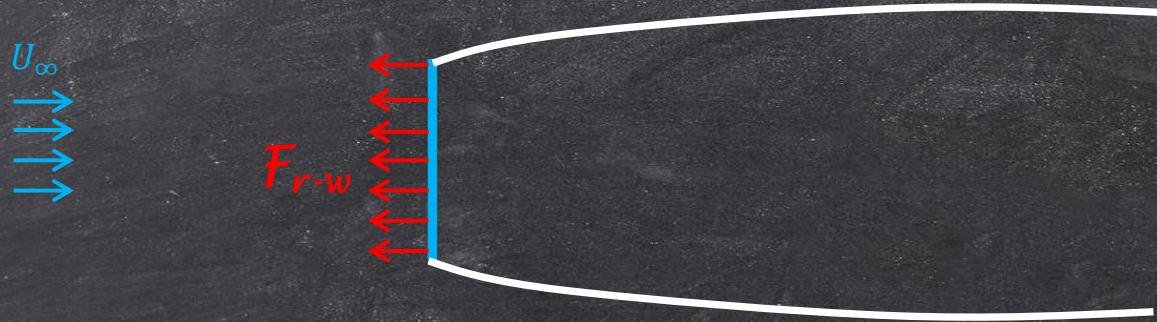


- The curl simplifies into: $\nabla \times f = \begin{bmatrix} \cancel{\frac{f_z}{dy}} - \cancel{\frac{f_y}{dz}} \\ -\cancel{\frac{f_x}{dx}} + \cancel{\frac{f_z}{dz}} \\ \cancel{\frac{f_y}{dx}} - \frac{df_x}{dy} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -\frac{df_x}{dy} \end{bmatrix}$

- The vorticity is only created only at the edges where $\frac{df_x}{dy}$ is not 0

Creation of Vorticity in a rotor

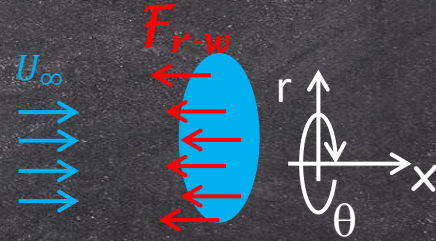
- Vortex sheet generation:



Creation of Vorticity in a rotor

- 3D actuator disk in cylindrical coordinates:

$$f = [f_r \ f_\theta \ f_x] = [0 \ 0 \ f_x]$$

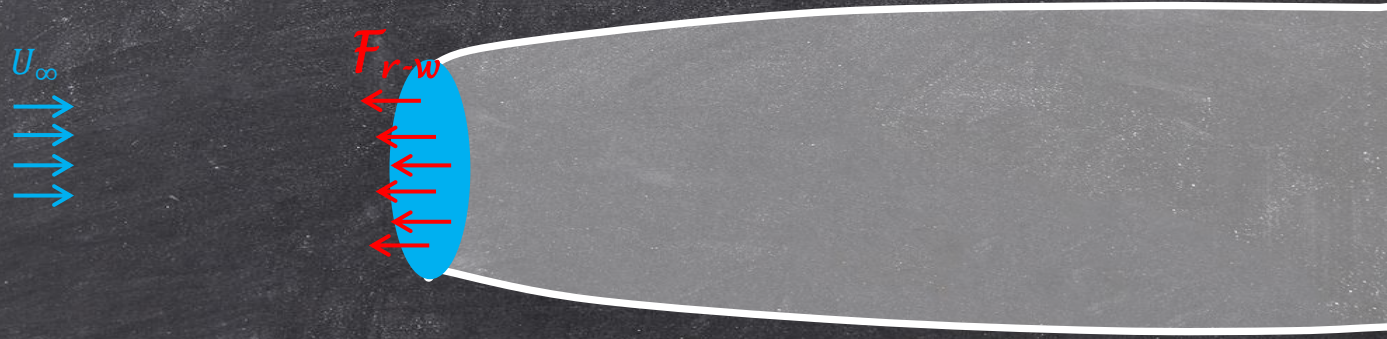


- Rewrite the curl $\nabla \times f = \begin{bmatrix} \frac{1}{r} \frac{\partial f_x}{\partial \theta} - \frac{\partial f_\theta}{\partial x} \\ -\frac{df_x}{dr} + \frac{\partial f_r}{\partial x} \\ \frac{\partial f_\theta}{dr} - \frac{1}{r} \frac{\partial f_r}{\partial \theta} \end{bmatrix} = \begin{bmatrix} 0 \\ -\frac{df_x}{dr} \\ 0 \end{bmatrix}$

- Also in the 3D case vorticity is only created at the edges

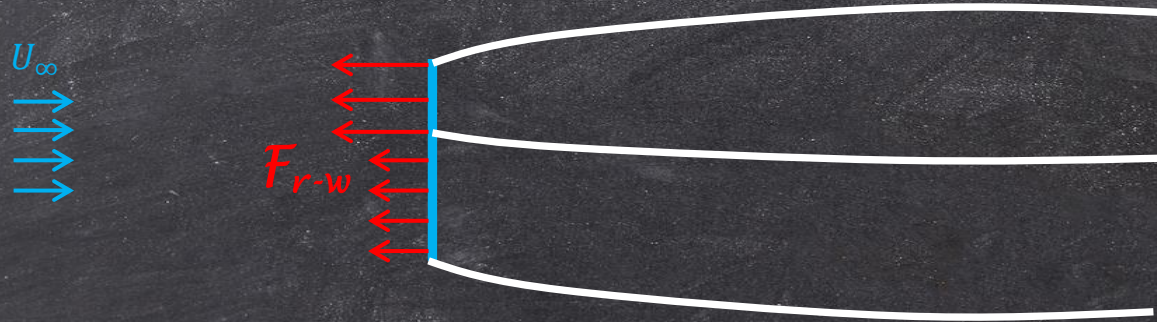
Creation of Vorticity in a rotor

- Vortex cylinder generation:

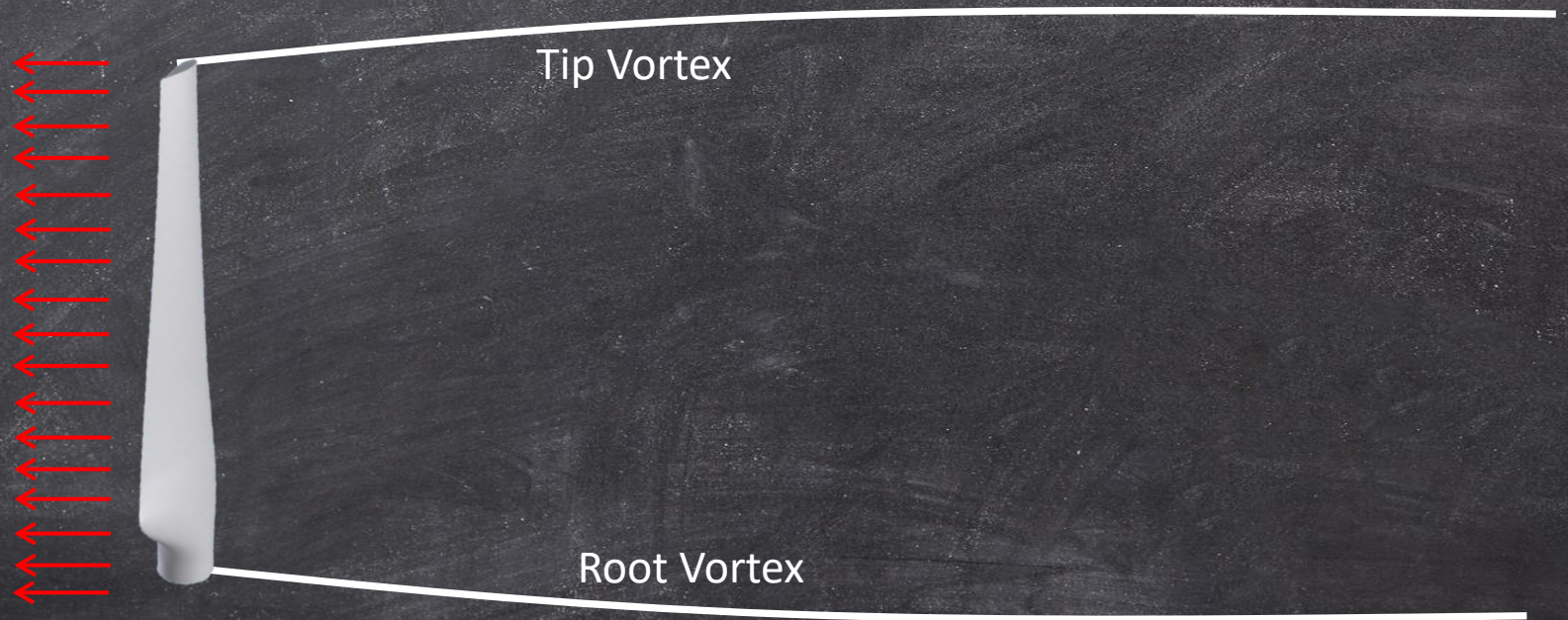


Creation of Vorticity in a rotor

- 2D actuator disk with non-uniform loading



Creation of Vorticity on a blade



Rotor and Wake Aerodynamics

Vorticity field generated from force field

Learning objectives

- To understand the process of vorticity generation given a space distribution of forces
 - An actuator disc case in 2D and 3D
 - Different actuator surfaces

Vorticity Equation

$$\frac{D\omega}{Dt} = \nabla \times \frac{Du}{Dt} = \frac{\partial \omega}{\partial t} + (u \cdot \nabla)\omega = \nabla \times f + (\omega \cdot \nabla)u + (\mu \nabla^2 \omega)$$

source/sink
vorticity turning and stretching
diffusion

- The material derivative of vorticity is composed of three terms:
1. A source/sink term, that is generated by a non-conservative force field ($\nabla \times f \neq 0$)
 2. A vorticity turning/stretching term $(\omega \cdot \nabla)u$, responsible for conserving the angular momentum.
 3. A vorticity diffusion term $\mu \nabla^2 \omega$.

Creation of Vorticity in a 2D AD

- In 2D, incompressible flow $(\omega \cdot \nabla)u = 0$, the vorticity equation is:

$$\frac{D\omega}{Dt} = \underbrace{\nabla \times f}_{\text{Source/sink}} + \underbrace{\mu \nabla^2 \omega}_{\text{diffusion}}$$

- This means that vorticity can only be introduced by a change in force field.

Creation of Vorticity in a 2D AD

➤ For an inviscid flow, the vorticity equation simplifies to

$$\frac{D\omega}{Dt} = \nabla \times \mathbf{f}$$

➤ Vorticity can only be generated by a curl in the force field $\mathbf{f}$.

$$\mathbf{f} = [f_x \ f_y \ f_z]$$

Creation of Vorticity in a 2D AD

$$f = [f_x \ f_y \ f_z]$$

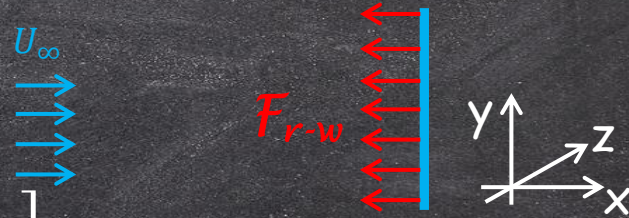
$$\nabla \times f = \begin{bmatrix} \frac{\partial f_z}{\partial y} - \frac{\partial f_y}{\partial z} \\ -\frac{\partial f_z}{\partial x} + \frac{\partial f_x}{\partial z} \\ \frac{\partial f_y}{\partial x} - \frac{\partial f_x}{\partial y} \end{bmatrix}$$

- Distribution of f in a 2D actuator disk with uniform loading



Creation of Vorticity in a 2D AD

- Force field $f = [f_x \ f_y \ f_z] = [f_x \ 0 \ 0]$

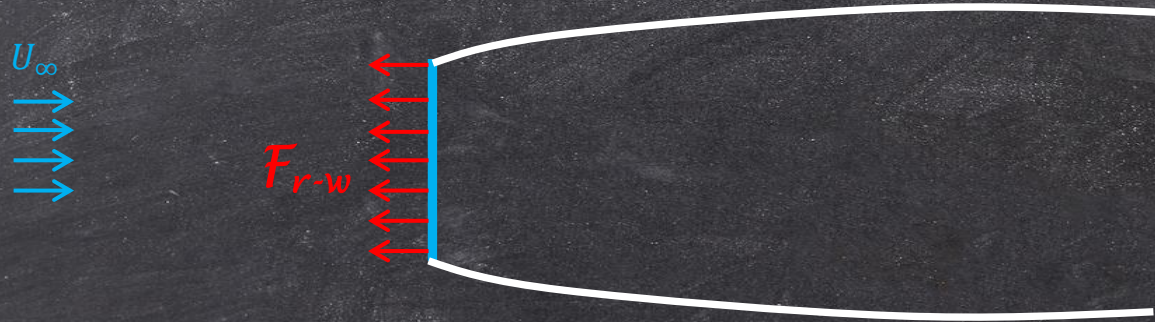


- The curl simplifies into: $\nabla \times f = \begin{bmatrix} \cancel{\frac{\partial f}{\partial y}} - \cancel{\frac{\partial f}{\partial z}} \\ -\cancel{\frac{\partial f}{\partial x}}z + \cancel{\frac{\partial f}{\partial x}} \\ \cancel{\frac{\partial f}{\partial x}} - \frac{\partial f_x}{\partial y} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ -\frac{\partial f_x}{\partial y} \end{bmatrix}$

- The vorticity is only created only at the edges where $\frac{\partial f_x}{\partial y}$ is not 0

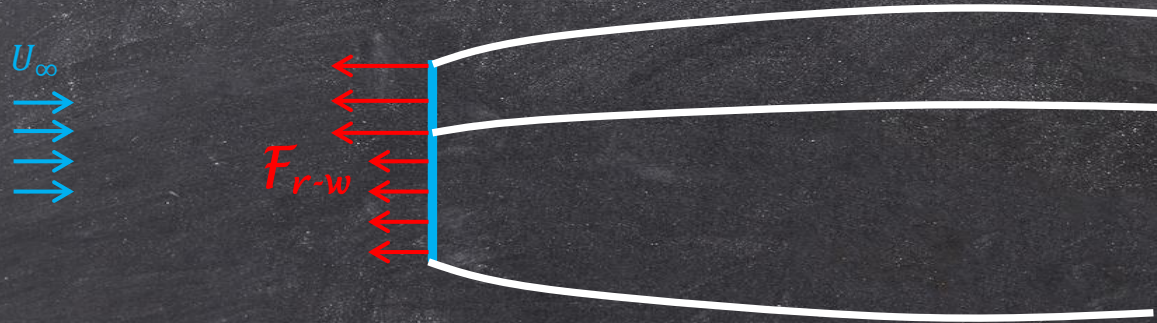
Creation of Vorticity in a 2D AD

- Vortex sheet generation:



Creation of Vorticity in a rotor

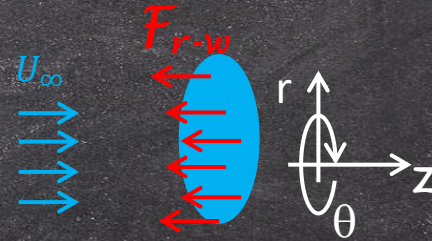
- 2D actuator disk with non-uniform loading



Creation of Vorticity in a 3D AD

- 3D actuator disk in cylindrical coordinates:

$$f = [f_r \ f_\theta \ f_z] = [0 \ 0 \ f]$$

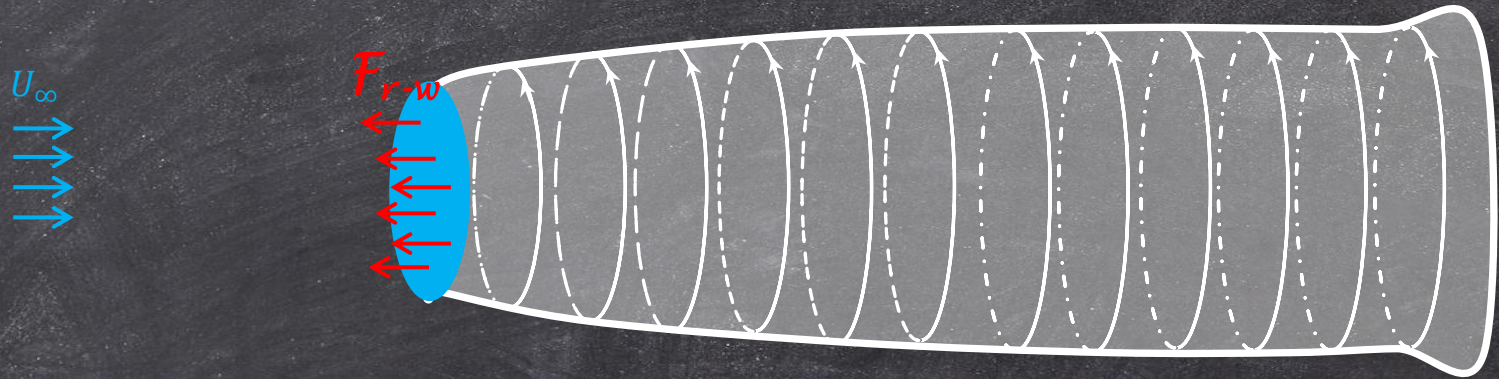


- Rewrite the curl $\nabla \times f = \begin{bmatrix} \frac{1}{r} \frac{\partial f_z}{\partial \theta} - \frac{\partial f_\theta}{\partial z} \\ -\frac{\partial f_z}{\partial r} + \frac{\partial f_r}{\partial z} \\ \frac{\partial f_\theta}{\partial r} - \frac{1}{r} \frac{\partial f_r}{\partial \theta} \end{bmatrix} = \begin{bmatrix} 0 \\ -\frac{\partial f_z}{\partial r} \\ 0 \end{bmatrix}$

- Also in the 3D case vorticity is only created at the edges

Creation of Vorticity in a 3D AD

- Vortex cylinder generation:



Today we:

- Demonstrated how the vorticity field is generated for a rotor (2D/3D actuator disc)
- Understand that to design a rotor is to design a desired force distribution, by sweeping a surface in space and time.

Rotor and Wake Aerodynamics

Stagnation enthalpy and total
pressure

Our objective for today

To demonstrate how the force field and the wake represent boundaries of energy as regions where the stagnation pressure of the flow changes:

- To derive the equation of stagnation pressure;
- To understand how the stagnation pressure changes with a force field and a vorticity field;
- To understand the wake of a rotor as a control volume whose boundary surfaces are discontinuities of the stagnation pressure.

The Stagnation enthalpy equation

Perfect gas $p = \rho R_g T$ $pv_g = R_g T$

R_g - the specific gas constant, $287 \text{ J}/(\text{kg} \cdot \text{K})$ at standard air conditions.

pv_g - the work necessary to establish the volume of flow at the given pressure per unit mass.

e - the specific internal energy (the internal energy per unit mass), representing the energy at molecular scale.

The combination of the two lead to the specific enthalpy h :

$$h = e + pv_g$$

The Stagnation enthalpy equation

Perfect gas, e and h are linear(calorically perfect fluid) function of T :

$$e = c_v T \quad h = c_p T$$

c_v and c_p are the specific heat at constant volume and constant pressure, which relate to the specific gas constant by Carnot's law:

$$c_p - c_v = R_g \quad c_p = \frac{\gamma R}{\gamma - 1} \quad c_v = \frac{R}{\gamma - 1} \quad \gamma = \frac{c_p}{c_v}$$

$\gamma = 1.4$ at standard air conditions

The Stagnation enthalpy equation

The specific total energy e_t and the specific total enthalpy h_t :

$$e_t = e + \frac{v^2}{2} \quad h_t = h + \frac{v^2}{2}$$

By using the second law of thermodynamics assuming small reversible changes, the material derivative of the specific energy and pressure can be written as

$$\frac{d}{dt} \left(e + \frac{p}{\rho} \right) = T \frac{ds_g}{dt} + \frac{1}{\rho} \frac{dp}{dt}$$

where s_g is the specific entropy. If the flow is assumed to be isentropic and incompressible, we have

$$\frac{de}{dt} = 0$$

The Stagnation enthalpy equation

The partial differential equation for the total energy of the flow in an incompressible, adiabatic and inviscid flow is

$$\rho \frac{\partial}{\partial t} \left(e + \frac{v^2}{2} \right) + \rho \vec{\nabla} \cdot \left[\left(e + \frac{v^2}{2} \right) \vec{v} \right] = -\vec{\nabla} \cdot (p \vec{v}) + \vec{f} \cdot \vec{v}$$

With isentropic flow assumption, the equation is simplified into

$$\rho \frac{\partial}{\partial t} \left(\frac{v^2}{2} \right) + \rho \vec{\nabla} \cdot \left[\left(\frac{v^2}{2} \right) \vec{v} \right] = -\vec{\nabla} \cdot (p \vec{v}) + \vec{f} \cdot \vec{v} \Rightarrow \rho \frac{\partial}{\partial t} \left(\frac{v^2}{2} \right) + \rho \vec{\nabla} \cdot \left[\left(\frac{v^2}{2} + p \right) \vec{v} \right] = \vec{f} \cdot \vec{v}$$

We will disregard the internal energy under these flow conditions and define the specific pressure or stagnation enthalpy as

$$h_s = \frac{p}{\rho} + \frac{v^2}{2}$$

The Stagnation enthalpy equation

To understand how the stagnation pressure varies along the flow, we take the momentum equation and combine it with vectorial identity and re-write it with a focus in the stagnation pressure:

$$\frac{\partial \vec{v}}{\partial t} + \vec{\nabla} \left(\frac{v^2}{2} + \frac{p}{\rho} \right) = \vec{v} \times (\vec{\nabla} \times \vec{v}) + \frac{\vec{f}}{\rho} + \nu \nabla^2 \vec{v} \Rightarrow \vec{\nabla} h_s = -\frac{\partial \vec{v}}{\partial t} + \vec{v} \times \vec{\omega} + \frac{\vec{f}}{\rho} + \nu \nabla^2 \vec{v}$$

The stagnation enthalpy can change in space due to :

1. acceleration of the flow field.
2. external force field applied to the flow.
3. the presence of a vorticity field.
4. the effect of viscosity.

Stagnation pressure test examples

- In these examples, we consider steady inviscid flow, the variation of stagnation pressure becomes

$$\vec{\nabla} h_s = \cancel{\frac{\partial \vec{h}}{\partial t}} + \vec{v} \times \vec{\omega} + \frac{\vec{f}}{\rho} + \cancel{\frac{\partial \vec{h}}{\partial t}}$$

- $\vec{v} \times \vec{\omega}$ term in cartesian and cylindrical coordinate system become

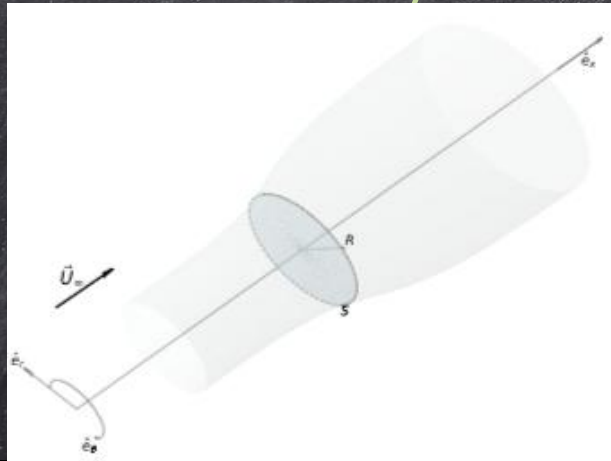
$$\vec{v} \times \vec{\omega} = \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix} \times \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} = \begin{vmatrix} \hat{e}_x & \hat{e}_y & \hat{e}_z \\ v_x & v_y & v_z \\ \omega_x & \omega_y & \omega_z \end{vmatrix} = \begin{bmatrix} v_y \omega_z - v_z \omega_y \\ -v_x \omega_z + v_z \omega_x \\ v_x \omega_y - v_y \omega_x \end{bmatrix}$$

$$\vec{v} \times \vec{\omega} = \begin{bmatrix} v_r \\ v_\theta \\ v_z \end{bmatrix} \times \begin{bmatrix} \omega_r \\ \omega_\theta \\ \omega_z \end{bmatrix} = \begin{vmatrix} \hat{e}_r & \hat{e}_\theta & \hat{e}_z \\ v_r & v_\theta & v_z \\ \omega_r & \omega_\theta & \omega_z \end{vmatrix} = \begin{bmatrix} v_\theta \omega_z - v_z \omega_\theta \\ -v_r \omega_z + v_z \omega_r \\ v_r \omega_\theta - v_\theta \omega_r \end{bmatrix}$$

Stagnation pressure variation across an AD

In this example, we will evaluate the change in h_s across an AD in cylindrical coordinate system for two scenarios:

1. A uniform force distribution in axial flow direction: $\vec{f} = [0; 0; f_z]$.
2. A force field in the azimuthal direction with a distribution varying inversely with the radius : $\vec{f} = [0; f_\theta; 0]$, where $|f_\theta| \propto \frac{1}{r}$.

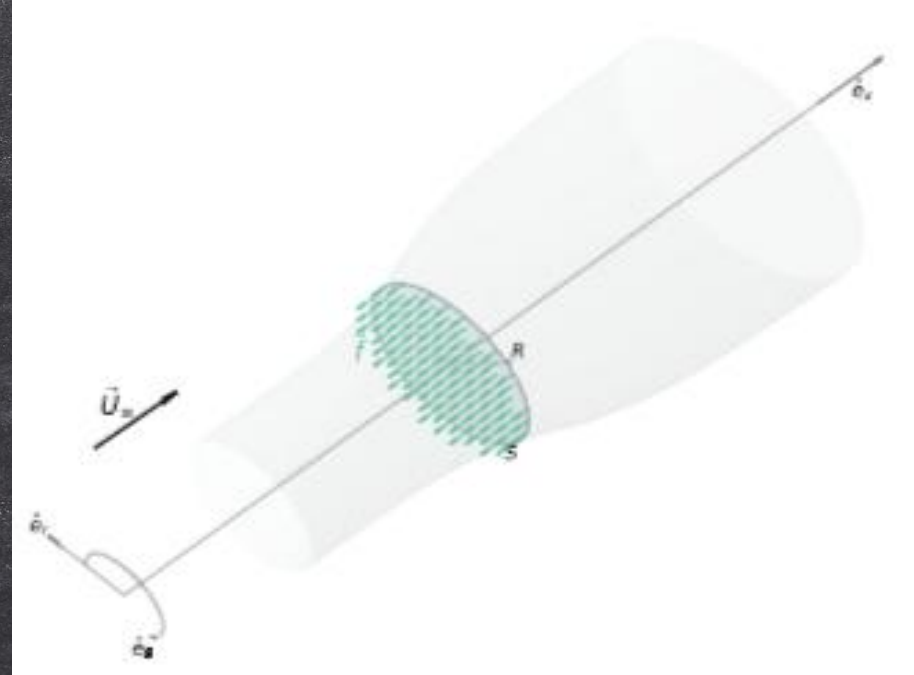


Representation of an AD of radius R perpendicular to the unperturbed flow velocity $\vec{U}_\infty$. The semi-transparent surface represents the boundary of the streamtube of the particles that cross the AD.

Stagnation pressure variation across an AD

1. A uniform force distribution in axial flow direction: $\vec{f} = [0; 0; f_z]$.
For steady, inviscid flow

$$\vec{\nabla} h_s = \vec{v} \times \vec{\omega} + \frac{\vec{f}}{\rho}$$



Stagnation pressure variation across an AD

flow direction: $\vec{f} = [0; 0; f_z]$

Vorticity eq determine the generated vorticity:

$$\frac{D\omega}{Dt} = \nabla \times \mathbf{f} = \frac{1}{\rho} \frac{1}{r} \begin{bmatrix} \hat{e}_r & r\hat{e}_\theta & \hat{e}_z \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial z} \\ f_r & rf_\theta & f_z \end{bmatrix} = \frac{1}{\rho} \begin{bmatrix} \frac{1}{r} \frac{\partial f_z}{\partial \theta} - \frac{1}{r} \frac{\partial rf_\theta}{\partial z} \\ -\frac{\partial f_z}{\partial r} + \frac{\partial f_r}{\partial z} \\ \frac{1}{r} \frac{\partial rf_\theta}{\partial r} - \frac{1}{r} \frac{\partial f_r}{\partial \theta} \end{bmatrix} = \begin{bmatrix} 0 \\ -\frac{\partial f_z}{\partial r} \\ 0 \end{bmatrix}$$

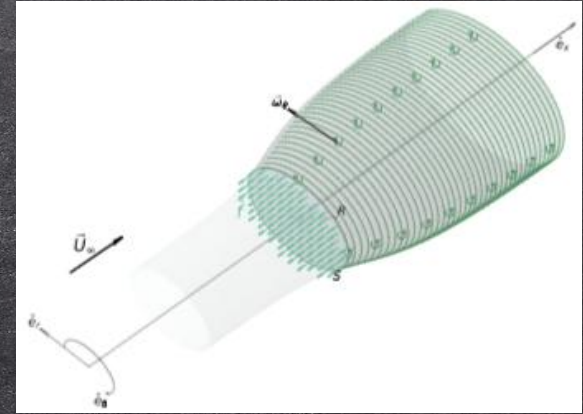
A uniform force distribution in axial flow direction: $\vec{f} = [0; 0; f_z]$

Because $\frac{\partial f_z}{\partial r}$ is only different from zero at AD edge, the actuator continuously generates vortex rings of azimuthal vorticity ω_θ , creating a vortex tube.

Stagnation pressure variation across an AD

If we consider this sheet to be infinitely thin, its strength at any point is defined by $\vec{\omega}$ with units of circulation per meter $\frac{m^2}{s^2} \frac{1}{m}$. As the wake is convected the strength of the sheet changes, but the modulus of the cross-product $|\vec{\omega} \times \vec{v}|$ remains constant along a streamline in this flow case. The wake is generated at the edge of AD;

we know previously that $\frac{D\omega}{Dt} = \begin{bmatrix} 0 \\ -\frac{\partial f_z}{\partial r} \\ 0 \end{bmatrix}$ we can derive that $|\vec{\omega} \times \vec{v}| = \left| \frac{f_z}{\rho} \right|$.



A schematic of the vorticity tube generated by an axial force field: $\vec{f} = [0; 0; f_z]$

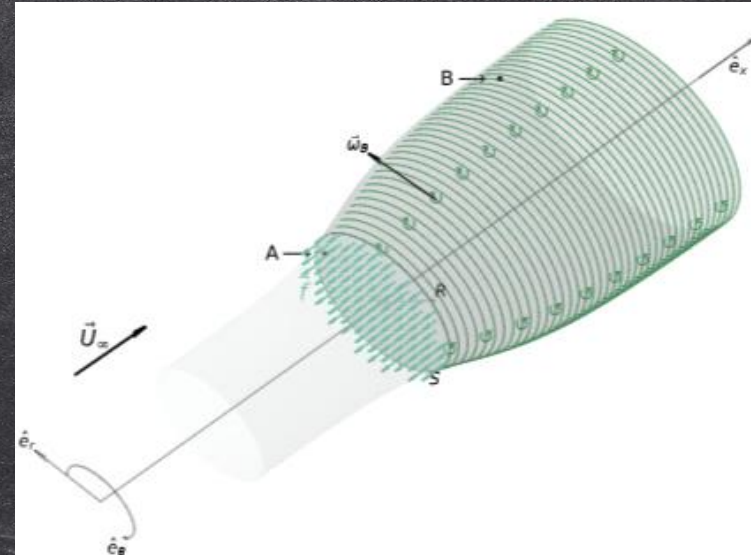
Stagnation pressure variation across an AD

Let's evaluate the change in h_s at : point A over the actuator surface and point B at the vortex sheet in the wake

At point A, $\vec{\omega} = 0$ and $\vec{f} = [0; 0; f_z]$,

$$\vec{\nabla} h_s = \vec{v} \times \vec{\omega} + \frac{\vec{f}}{\rho} = \frac{1}{\rho} \begin{bmatrix} 0 \\ 0 \\ f_z \end{bmatrix}$$

The stagnation pressure only changes in the axial direction, only determined by the force.



A schematic of the vortex tube generated by an axial force field: $\vec{f} = [0; 0; f_z]$

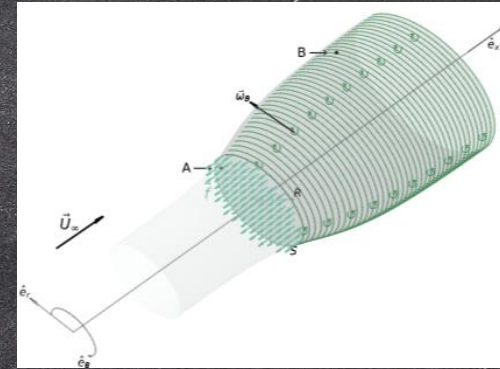
Stagnation pressure variation across an AD

At point B, $\vec{\omega} = [0; \omega_\theta; 0]$ and $\vec{f} = [0; 0; 0]$,

$$\vec{\nabla} h_s = \vec{v} \times \vec{\omega} + \frac{\vec{f}}{\rho} = \begin{bmatrix} v_z \omega_\theta - v_\theta \omega_z \\ -v_\theta \omega_r + v_r \omega_\theta \\ v_r \omega_\theta - v_\theta \omega_r \end{bmatrix} = \begin{bmatrix} -v_z \omega_\theta \\ 0 \\ v_r \omega_\theta \end{bmatrix}$$

Combining with $|\vec{\omega} \times \vec{v}| = \left| \frac{f_z}{\rho} \right|$, it gives $\begin{bmatrix} -v_z \omega_\theta \\ 0 \\ v_r \omega_\theta \end{bmatrix} = \left| \frac{f_z}{\rho} \right| \hat{e}_n$,

where $\hat{e}_n$ is the direction normal to the surface of the vortex sheet at point B. As we expected, the jump of h_s across the AD is the same as across the vortex sheet that bounds the wake. The wake is therefore boundary between regions with different total/stagnation pressure. Any attempt to "destroy" the vorticity sheet implies adding/subtracting energy to the wake.

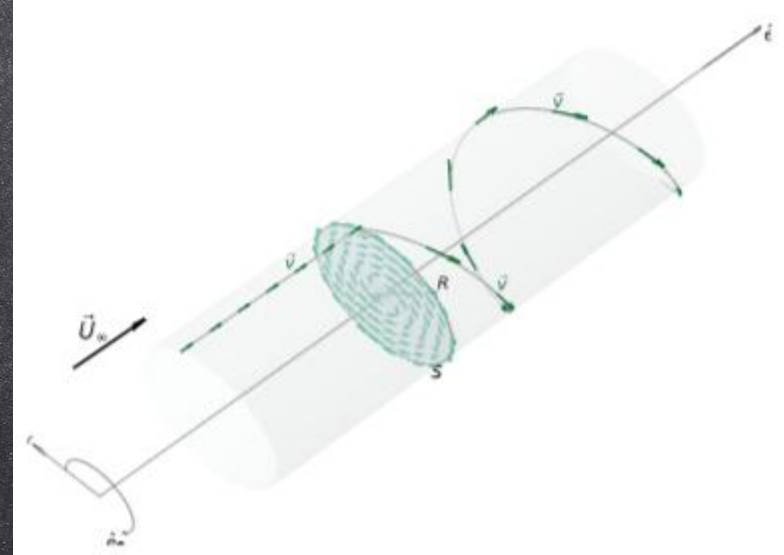


A schematic of the vortex tube generated by an axial force field $\vec{f} = [0; 0; f_z]$.

Stagnation pressure variation across an AD

Exercise 1 in class:

Scenario 2 : A force field in the azimuthal direction with a distribution varying inversely with the radius : $\vec{f} = [0; f_\theta; 0]$, where $|f_\theta| \propto \frac{1}{r}$.



A schematic of an AD with a force field in azimuthal flow direction θ , with a distribution of $f_\theta = \frac{f_0}{r}$

Stagnation pressure variation across an AD

Scenario 2 : A force field in the azimuthal direction with a distribution varying inversely with the radius : $\vec{f} = [0; f_\theta; 0]$, where $f_\theta = \frac{f_0}{r}$.

Similarly, for the vorticity generation at the AD:

$$\frac{D\omega}{Dt} = \nabla \times \mathbf{f} = \frac{1}{\rho} \begin{bmatrix} \cancel{\frac{1}{r} \frac{\partial f_z}{\partial \theta}} - \frac{1}{r} \frac{\partial r f_\theta}{\partial z} \\ -\cancel{\frac{\partial f}{\partial r}} + \cancel{\frac{\partial f}{\partial z}} \\ \frac{1}{r} \frac{\partial r f_\theta}{\partial r} - \cancel{\frac{1}{r} \frac{\partial f}{\partial \theta}} \end{bmatrix} = \frac{1}{\rho} \begin{bmatrix} -\frac{1}{r} \frac{\partial r f_\theta}{\partial z} \\ 0 \\ \frac{1}{r} \frac{\partial r f_\theta}{\partial r} \end{bmatrix} = \frac{1}{\rho} \begin{bmatrix} -\frac{1}{r} \frac{\partial f_0}{\partial z} \\ 0 \\ \frac{1}{r} \frac{\partial f_0}{\partial r} \end{bmatrix}$$

We will now evaluate the above eq for three different type of streamlines:

1. A streamline that crosses the AD at the root $r=0$.
2. A streamline over the boundary of the streamtube that passes the edge of AD at $r=R$.
3. A streamline that crosses the AD at $0 < r < R$.

Stagnation pressure variation across an AD

Scenario 2 : An azimuthal force field: $\vec{f} = [0; f_\theta; 0]$, where $f_\theta = \frac{f_0}{r}$.

At the streamline across $r=0$, we have $\frac{d\vec{\omega}}{dt} = [0; 0; \frac{f_0}{r}]$. The solution of the singularity is the generation of a concentrated vortex at the root with strength $\frac{2\pi f_0}{\rho U_\infty}$ in the direction $\hat{e}_z$. Because the AD is infinitely thin, f_0 is a delta function and has a unit of N/m.

At the streamline across $r=R$, we have $\frac{d\vec{\omega}}{dt} = [0; 0; \frac{f_0}{R}]$. Vortex lines are shed with strength $\frac{f_0}{\rho U_\infty R}$. Despite the helical shape of the streamline, because the tangential velocity is constant, the vorticity remains aligned with $\hat{e}_z$.

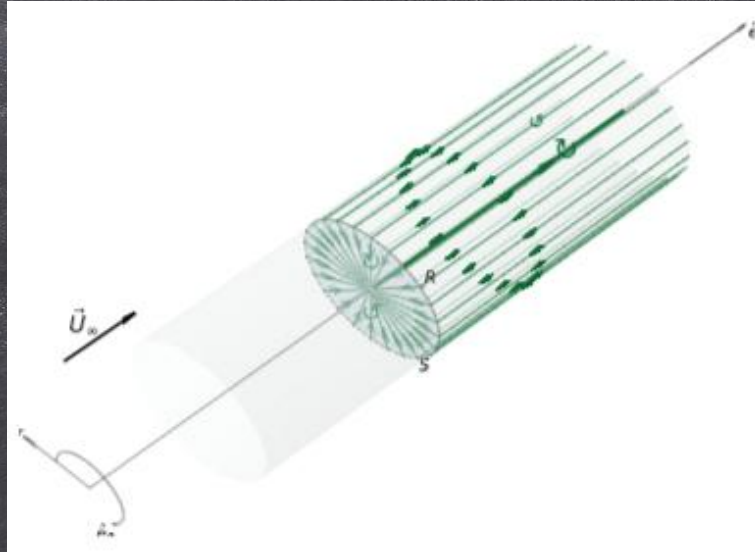
Stagnation pressure variation across an AD

Scenario 2 : An azimuthal force field: $\vec{f} = [0; f_\theta; 0]$, where $f_\theta = \frac{f_0}{r}$.

At the streamline crosses the actuator at $0 < r < R$, we have again the effect of the delta function, where vorticity is created and destroyed as the flow elements crosses the actuator. On the upwind side of the actuator $\frac{d\vec{\omega}}{dt} = \left[-\frac{f_0}{r}; 0; 0\right]$ and on the downwind side of the actuator $\frac{d\vec{\omega}}{dt} = \left[\frac{f_0}{r}; 0; 0\right]$. At the actuator there is a radial vorticity sheet with strength $-\frac{f_0}{\rho U_\infty r}$.

Stagnation pressure variation across an AD

Scenario 2 : An azimuthal force field: $\vec{f} = [0; f_\theta; 0]$, where $f_\theta = \frac{f_0}{r}$.



A schematic of vorticity field of an AD generated with a force field in azimuthal flow direction θ , with a distribution of $f_\theta = \frac{f_0}{r}$

What we have achieved:

We have learnt how the force field and the wake represent boundaries of energy as regions where the stagnation pressure of the flow changes.

We

- understand the derivation of the stagnation pressure.
- understand how the stagnation pressure changes with a force field and a vorticity field using the examples of various actuator surfaces.
- understand the wake of a rotor as a control volume whose boundary surfaces are discontinuities of the stagnation pressure.

Rotor and Wake Aerodynamics

Derivation of the potential flow equations

Our objective for today:

- To derive a lagrangian potential flow model
 - Obtain a linear model that allows us to calculate velocity and pressure by combination elementary flow solutions
- Define elementary solutions and how to combine them

The physical meaning of rotation

The rotation of a solid body

$$\vec{V} = \vec{V}_0 + \vec{\Omega} \times \vec{r}$$



$$v_x = v_{0x} + \Omega_y z - \Omega_z y$$

$$v_y = v_{0y} + \Omega_z x - \Omega_x z$$

$$v_z = v_{0z} + \Omega_x y - \Omega_y x$$

The physical meaning of rotation

The curl of the velocity field is equal to the vorticity

$$\begin{aligned}\vec{\omega} &= \vec{\nabla} \times \vec{v} \\ &= \left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z} \right) \vec{i} + \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x} \right) \vec{j} + \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y} \right) \vec{k} \\ &= 2\Omega_x \vec{i} + 2\Omega_y \vec{j} + 2\Omega_z \vec{k}\end{aligned}$$

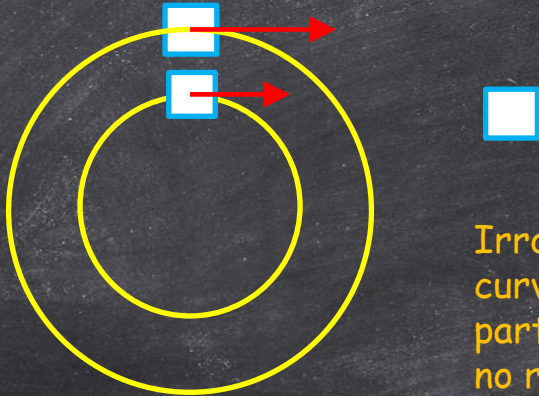
Vorticity is twice the angular velocity of the flow particles

$$\vec{\omega} = 2\vec{\Omega}$$

Rotational and Irrotational flow

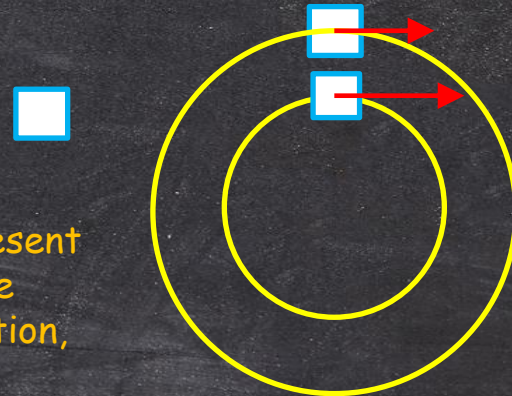
$$\vec{\omega} \neq 0$$

The flow particles have a finite rotational velocity, the flow is rotational



$$\vec{\omega} = 0$$

The flow particles have no rotational velocity, the flow is irrotational



Irrotational flow can represent curved streamlines, but the particles only have translation, no rotation

Velocity potential

$$\vec{\nabla} \times \vec{V} = 0$$

Irrotational flow

$$\vec{\nabla} \times (\vec{\nabla} \Phi) = 0$$

if Φ is a scalar function

$$\vec{V} = \vec{\nabla} \Phi$$

For irrotational flow, we can define a scalar function Φ , such that its gradient is the velocity field. Φ is the velocity potential.

Velocity potential is a linear function

$$\vec{\nabla} \cdot \vec{V} = 0 \quad \text{Incompressible flow}$$

$$\vec{\nabla} \cdot (\vec{\nabla} \Phi) = \nabla^2 \Phi = 0 \quad \text{which is Laplace's equation}$$

Solutions of the Laplace equation are linear. We can create a velocity potential by linear superposition of individual velocity potentials

$$\Phi = \Phi_1 + \Phi_2 + \dots + \Phi_n$$

$$\vec{V} = \vec{V}_1 + \vec{V}_2 + \dots + \vec{V}_n$$

Creation of a potential flow model

- Define elementary flow solutions
- Define boundary conditions
- Solve Laplace's equation
- Calculate velocity
- Calculate pressure using the Bernoulli equation

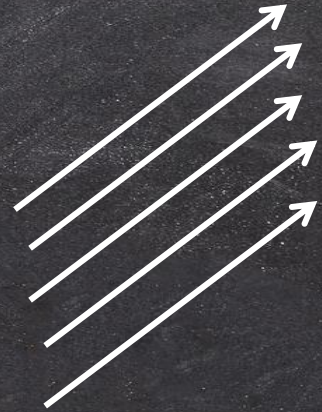
Elementary flow solution #1

Uniform flow

$$\Phi = U_{\infty}x + V_{\infty}y + W_{\infty}z \quad \text{Uniform flow potential}$$

$$\vec{V} = \begin{bmatrix} u \\ v \\ w \end{bmatrix} = \vec{\nabla}\Phi = \begin{bmatrix} U_{\infty} \\ V_{\infty} \\ W_{\infty} \end{bmatrix} \quad \text{Uniform flow velocity field}$$

$$\vec{\Gamma} = - \iint_S \vec{\nabla} \times \vec{V} dS = 0 \quad \text{Circulation is null}$$



Elementary flow solution #2

Source flow

$$\Phi = \frac{\Lambda}{2\pi} \ln(r)$$

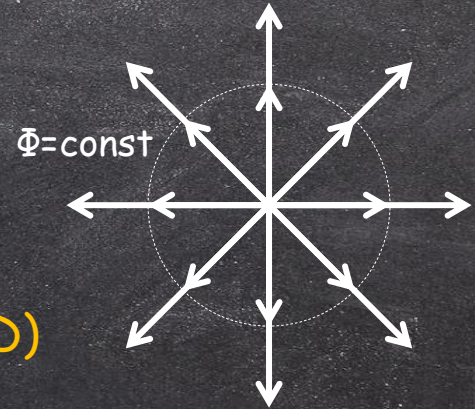
Source flow potential (2D)

$$\vec{V}_r = \frac{\Lambda}{2\pi r}$$

Source flow velocity field (2D)

$$\vec{\Gamma} = - \iint_S \vec{\nabla} \times \vec{V} dS = 0$$

Circulation is null



Elementary flow solution #3

Doublet flow

$$\Phi = \frac{\kappa}{2\pi} \frac{\cos\theta}{r}$$

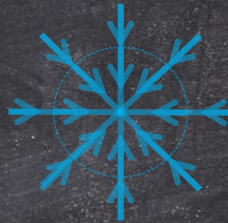
Source flow potential (2D)

$$\vec{V}_r = -\frac{\kappa}{2\pi} \frac{\cos\theta}{r^2}$$

Source flow velocity field (2D)

$$\vec{\Gamma} = -\iint_S \vec{\nabla} \times \vec{V} dS = 0$$

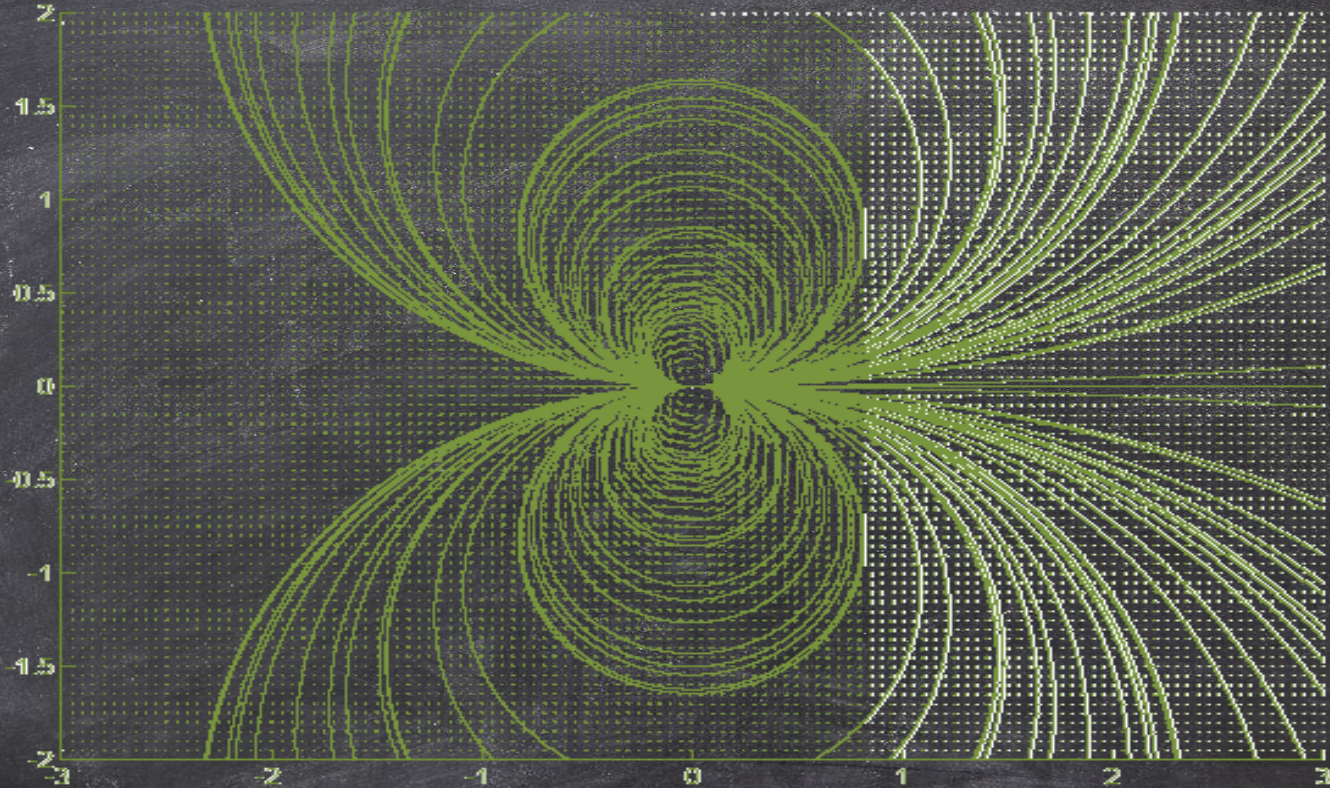
Circulation is null



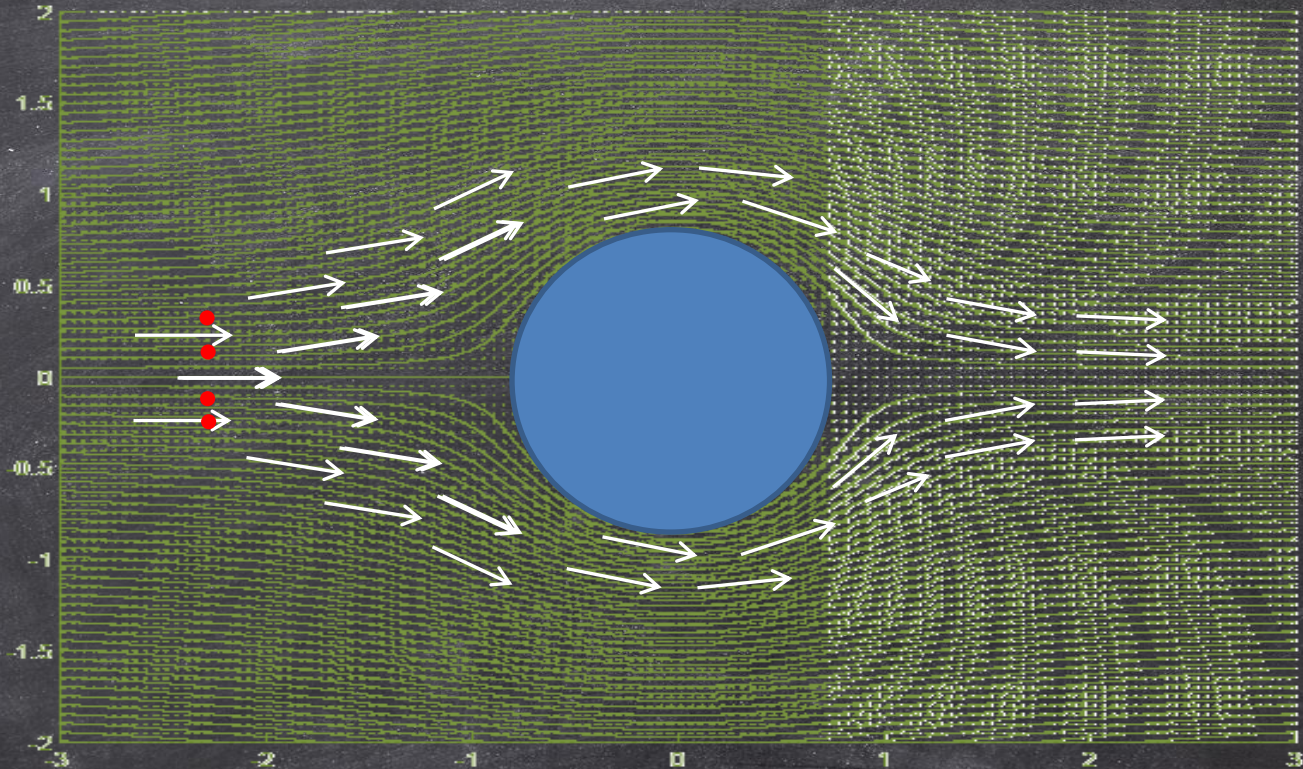
Source-sink pair infinitely close

$$\kappa = l^* \Lambda$$

Doublet flow



Doublet in a uniform flow



Boundary and infinity conditions

Two conditions must be set:

$$\vec{u}_{wall} = \vec{v} \cdot n_{wall} = \nabla \Phi_{wall} \cdot n_{wall} = 0$$

non-permeability of surface (no flow through the walls of the body)

$$\nabla \Phi_{\infty} = \vec{v}_{\infty}$$

regularity at infinity

Elementary flow solution #4

Vortex flow

$$\Phi = -\frac{\Gamma}{2\pi}\theta$$

Vortex flow potential (2D)

$$\vec{V}_r = 0; \vec{V}_\theta = -\frac{\Gamma}{2\pi r}$$

Vortex flow velocity field (2D)

Singularity for $r \rightarrow 0$!

$$\vec{\Gamma} = -\iint_S \vec{\nabla} \times \vec{V} dS = \vec{\Gamma} \quad \text{Circulation is } \neq 0$$



Free vortex, forced vortex and real vortex (I)

Suppose a flow field with circular trajectories

$$\vec{v} = [v_r v_\theta v_x], v_r = v_x = 0, v_\theta \neq 0$$

The vorticity is then

$$\vec{\omega} = \nabla \times \vec{v} = \left[0, 0, \frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right]$$

If the flow is irrotational, then $\omega = 0$, implying

$$\frac{\partial}{\partial r} (r v_\theta) = 0 \Rightarrow r v_\theta = \text{const}$$



Using singularities to create vorticity

Viscous core

To solve "the inconsistencies" of the Free Vortex, it is necessary to remove a singularity at the core of the vortex, such that

$$\lim_{ds \rightarrow 0} \omega ds = \Gamma$$

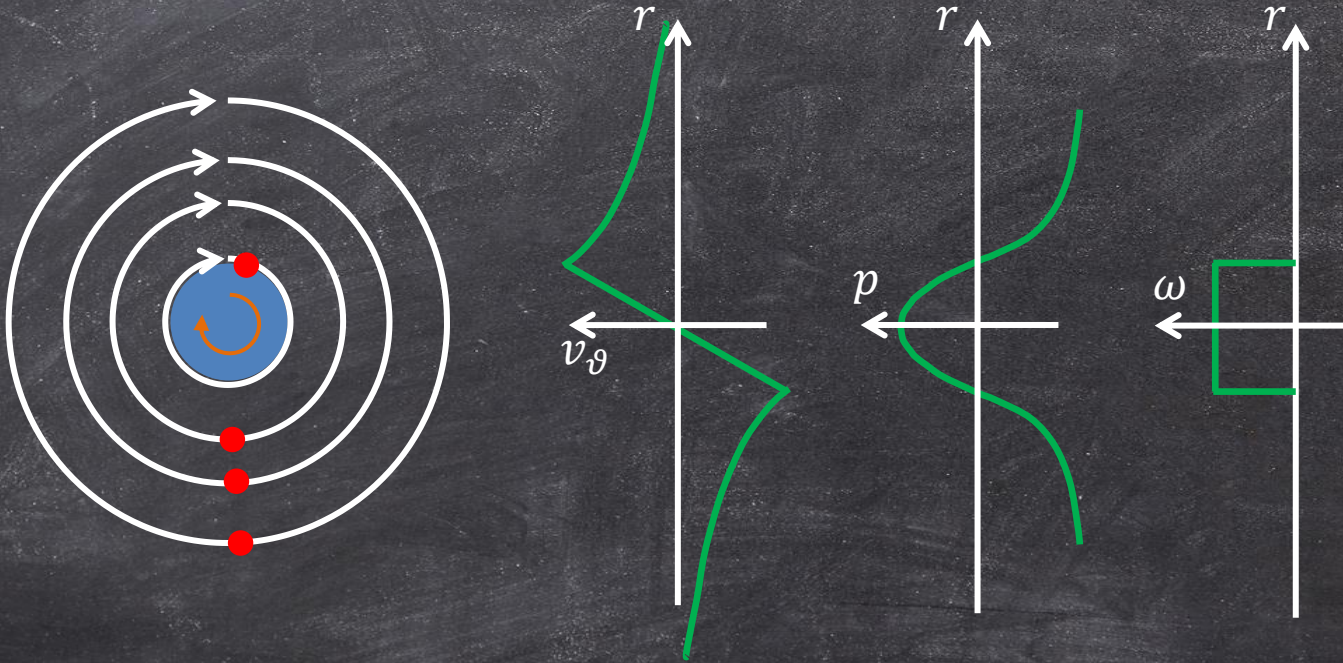
The core is then a body in solid rotation

$$v_\theta = r\Omega = r \frac{\omega}{2}$$



"Real" Rankine vortex (2D)

The Real Vortex is a combination of a Free Vortex and a viscous core

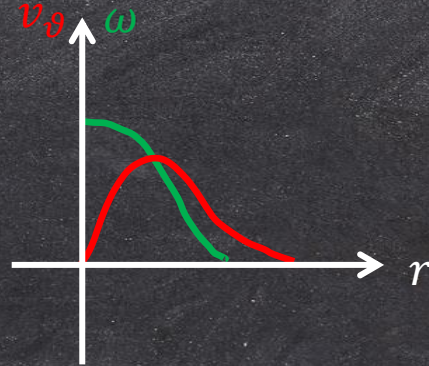


"Real" (Rankine) Vortex and Real Vortex

"Real" Rankine Vortex

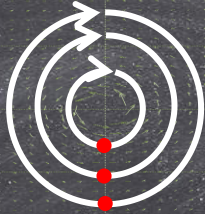


Real Vortex



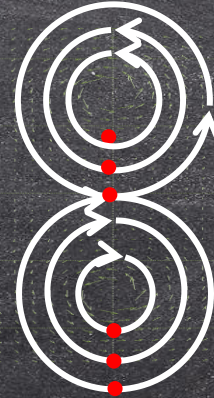
Superposing the solutions

Vortex

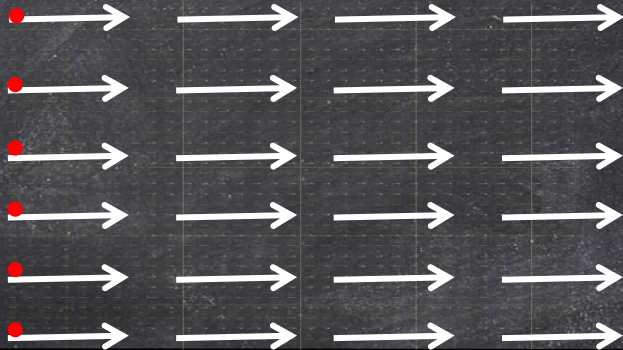


+

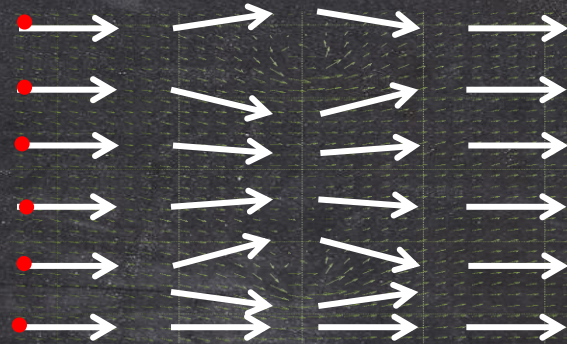
2 Vortices



Unperturbed flow



2 Vortices in unperturbed flow



Today, we

- Defined what is a potential flow model, and the condition of irrotationality of the flow
- Identified elementary flow solutions and combined them to create complex flows

Rotor and Wake Aerodynamics

Circulation and Helmholtz's Theorems

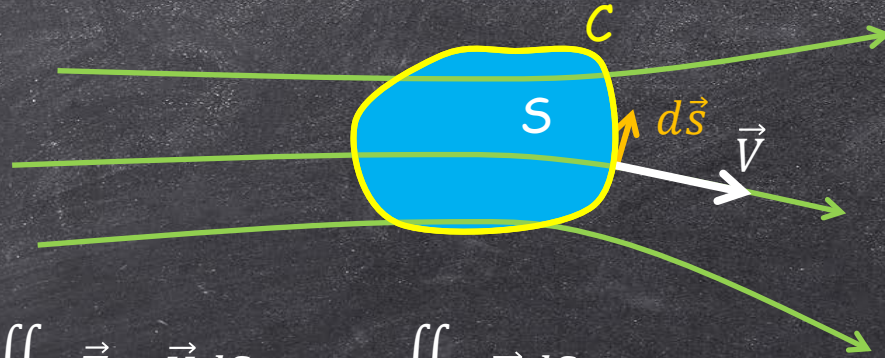


Learning objectives

- Define circulation
- Define Helmholtz' three theorems for the circulation
- Interpret Helmholtz' theorems as an application of Kelvin-Stokes theorem
- Infer the implication of these theorems on the modelling of vortices



Circulation



$$\vec{\Gamma} = - \oint_C \vec{V} \cdot d\vec{s} = - \iint_S \vec{\nabla} \times \vec{V} dS = - \iint_S \vec{\omega} dS$$

Circulation is equal to the vorticity integral over any open surface S bounded by C or to the contour integral along C of the velocity component tangential to the contour

Hermann von Helmholtz

1821-1894



Helmholtz's Theorems

Helmholtz Theorem 1: The strength of a vortex tube, which is defined as the circularity of the circumference of any cross section of the tube, is constant along its length. This theorem holds for any fluid, and simply derives from the fact that the vorticity field is divergence-free

Helmholtz Theorem 2: A vortex line cannot end in a fluid. Vortex lines and tubes must appear as closed loops, extend to infinity, or start/end at solid boundaries.

Helmholtz Theorem 3: A barotropic (or incompressible), inviscid fluid that is initially irrotational will remain irrotational. Hence, such a fluid does not and cannot create vorticity



Helmholtz's Theorems

Helmholtz Theorem 1: The strength of a vortex tube, which is defined as the circularity of the circumference of any cross section of the tube, is constant along its length. This theorem holds for any fluid, and simply derives from the fact that the vorticity field is divergence-free

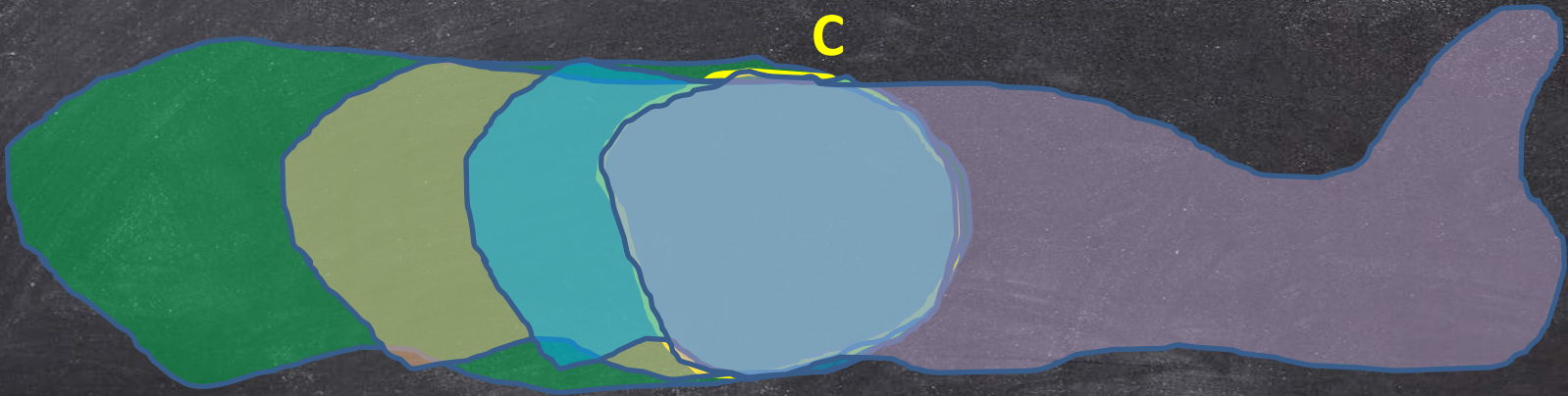
Helmholtz Theorem 2: A vortex line cannot end in a fluid. Vortex lines and tubes must appear as closed loops, extend to infinity, or start/end at solid boundaries.

Helmholtz Theorem 3: A barotropic (or incompressible), inviscid fluid that is initially irrotational will remain irrotational. Hence, such a fluid does not and cannot create vorticity



Helmholtz's Theorem 2

- Consider this contour C
- In 3D this contour bounds an infinite amount of surfaces



Helmholtz's Theorem 2

- Consider this contour C
- In 3D this contour bounds an infinite amount of surfaces
 - The choice of the surface for the surface integral cannot matter



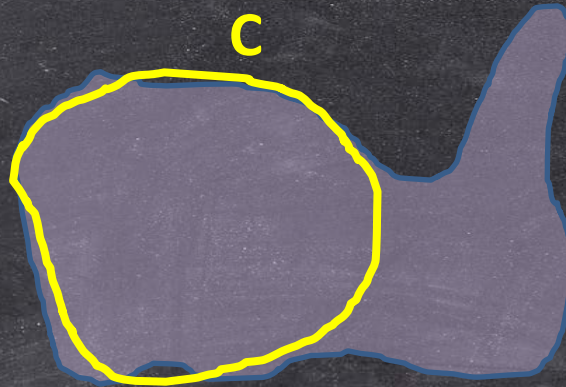
$$\vec{\Gamma} = - \oint_C \vec{V} \cdot d\vec{s}$$

$$\vec{\Gamma} = - \iint_S \vec{\omega} dS$$



Helmholtz's Theorem 2

- Consider this contour C
- In 3D this contour bounds an infinite amount of surfaces
 - The choice of the surface for the surface integral cannot matter



$$\vec{\Gamma} = - \oint_C \vec{V} \cdot d\vec{s}$$

$$\vec{\Gamma} = - \iint_S \vec{\omega} dS$$



Helmholtz's Theorem 2

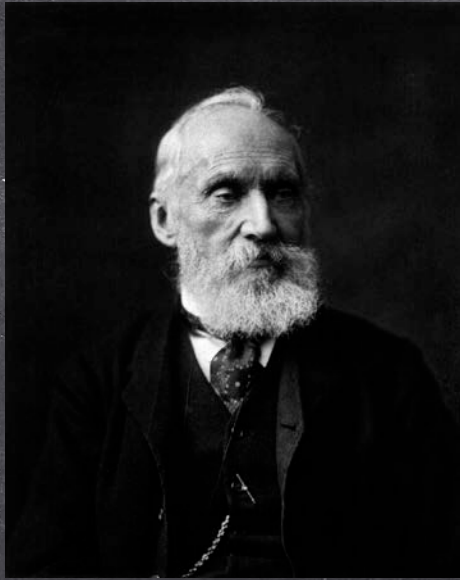
- A line of vorticity entering the surface cannot stop somewhere in space, because the integral would change depending on the surface.



Helmholtz's Theorem 2

- The vortex line has exit it again and close, leaving the sum of vorticity unchanged





Lord Kelvin
1824-1907



Sir Stokes
1819-1903



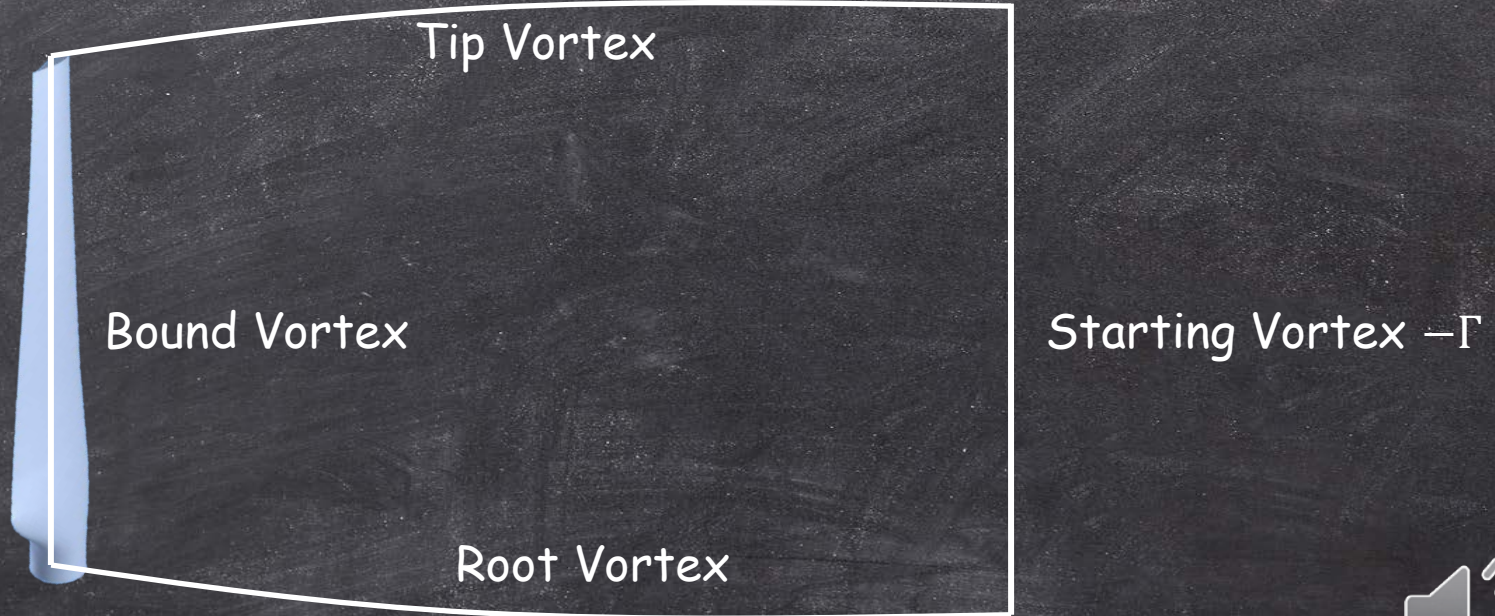
(Kelvin-) Stokes' theorem

$$\iint_S \nabla \times \vec{U} \cdot d\vec{S} = \iint_S \vec{\omega} \cdot d\vec{S} = \oint_C \vec{U} \cdot d\vec{l}$$



Circulation on a blade

Helmholtz Theorem 2



Today, we have...

- Defined circulation
- Defined Helmholtz' three theorems for the circulation
- Interpreted Helmholtz' theorems as an application of Kelvin-Stokes theorem
- Inferred the implication of these theorems on the modelling of vortices



Rotor and Wake Aerodynamics

Source/Sink flow example

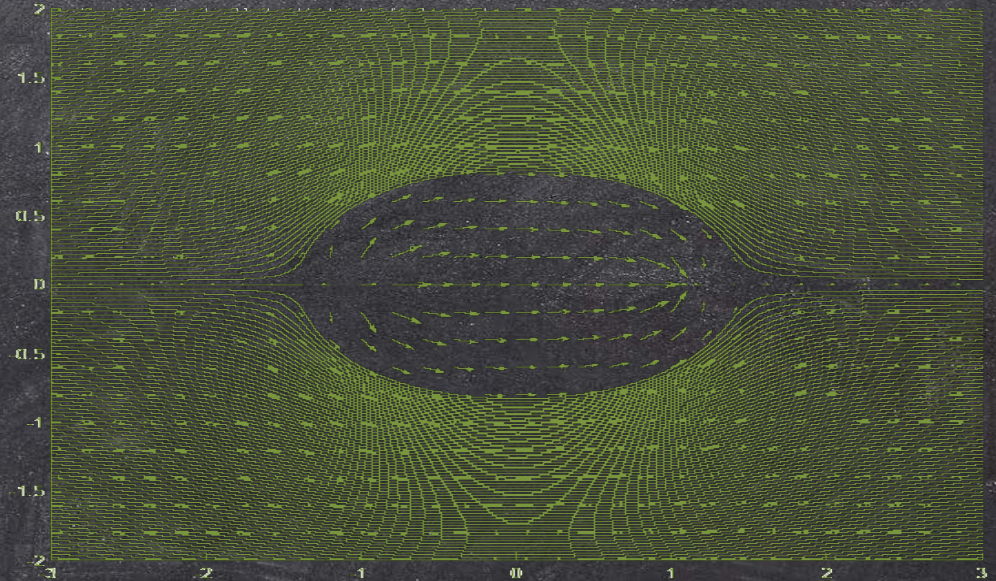
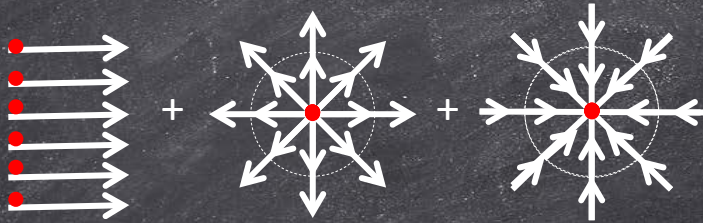


Learning objectives

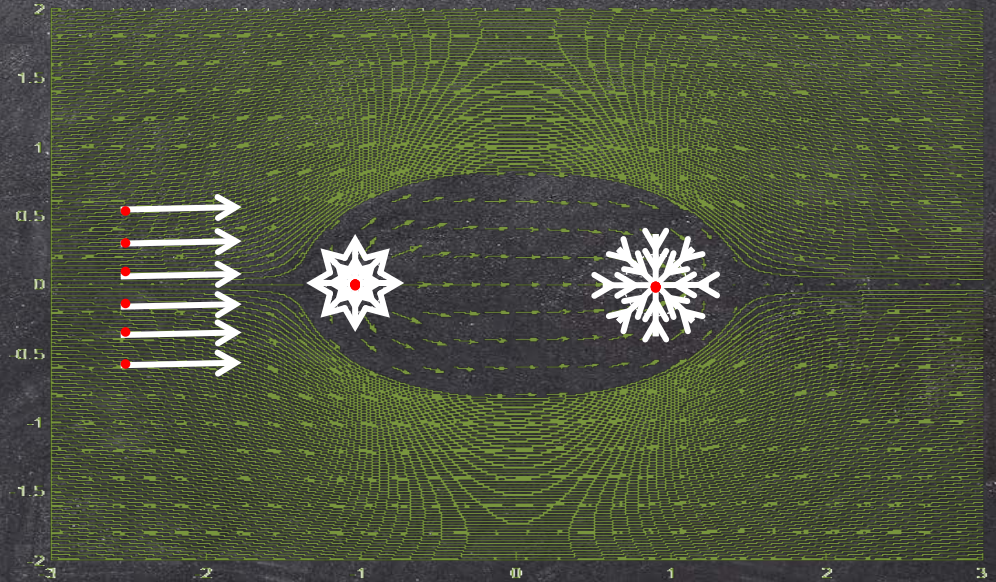
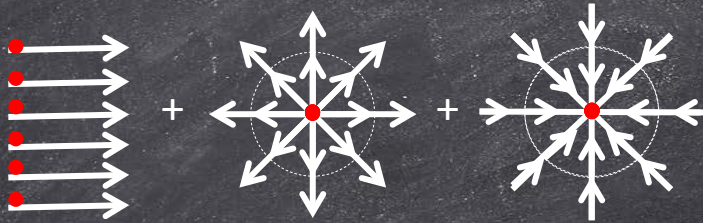
- To practice the concept of linearity of the potential flow equations
- To construct a solution of a flow field combining different elements
- To define the strength of the elements by setting a boundary condition



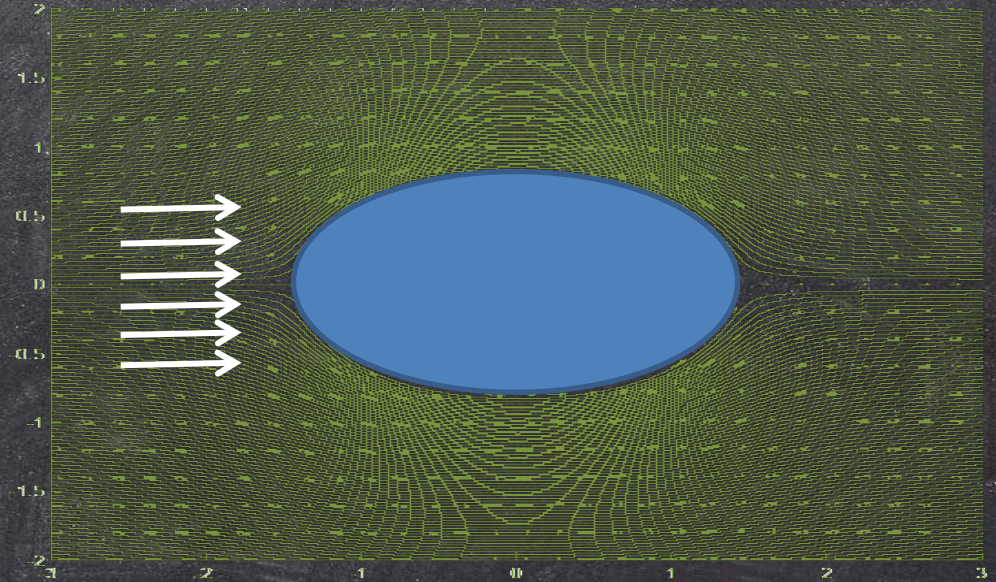
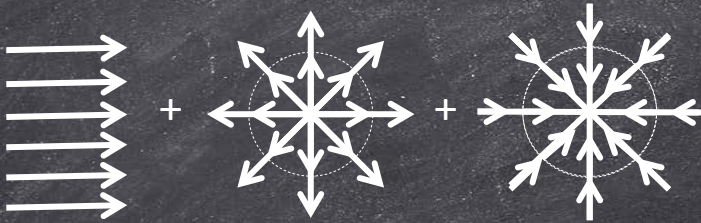
Source/sink pair - Rankine's oval



Source/sink pair - Rankine's oval



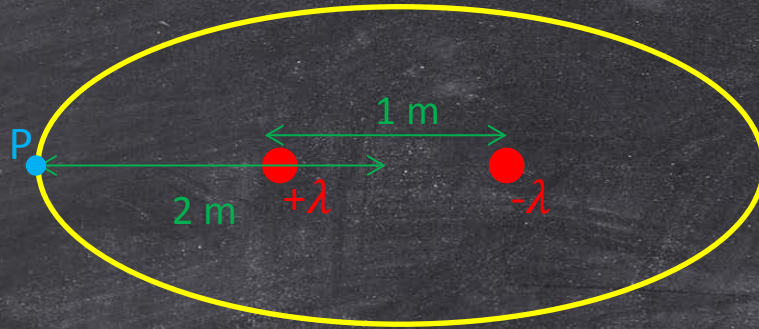
Source/sink pair - Rankine's oval



Flow over Rankine's oval

- How can we use source/sink pairs to model bodies in a flow?
- Consider this situation (source is indicated with $+\lambda$, sink with $-\lambda$):

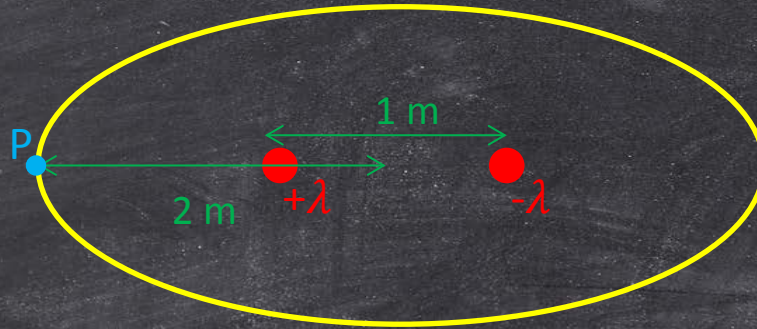
$$U_{\infty} = 1 \text{ m/s}$$



Flow over Rankine's oval

- We know that for a symmetric body, point P will be the stagnation point, which means that $U_P = 0$.
- So we sum up the contributions of all elementary flows at point P to find the velocity at P

$$U_\infty = 1 \text{ m/s}$$

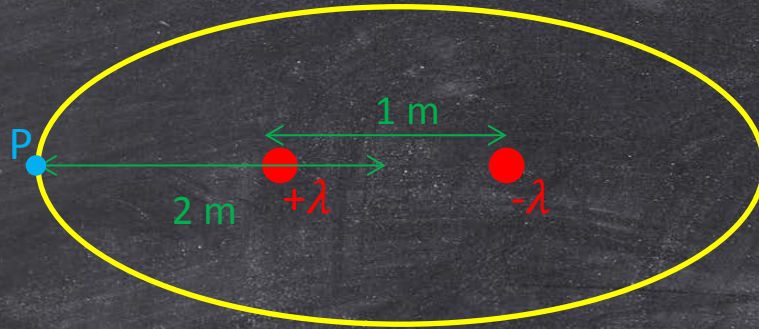


Flow over Rankine's oval

- We know that for a symmetric body, point P will be the stagnation point, which means that $U_P = 0$.

$$U_P = U_\infty(P) + U_{source}(P) + U_{sink}(P)$$

$$U_\infty = 1 \text{ m/s}$$

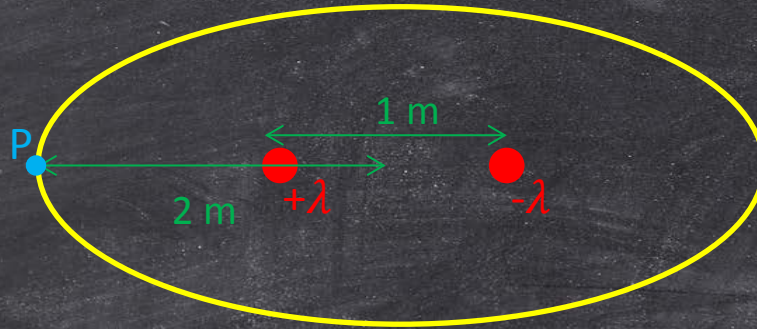


Flow over Rankine's oval

- We know that for a symmetric body, point P will be the stagnation point, which means that $U_P = 0$.

$$0 = 1 + \frac{\lambda}{2\pi(1.5)} + \frac{-\lambda}{2\pi(2.5)}$$

$$U_\infty = 1 \text{ m/s}$$



λ are equal because they must cancel each other to have no net effect on the flow!

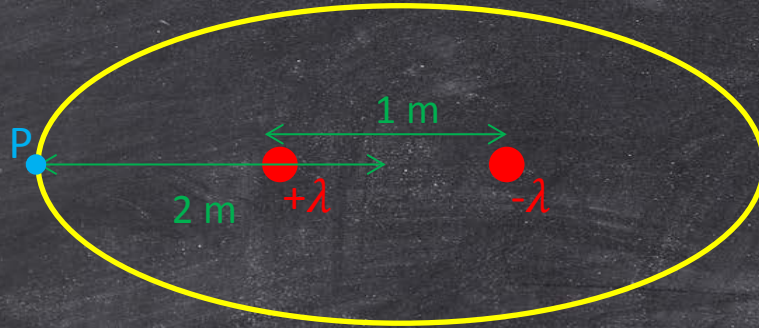


Flow over Rankine's oval

- We know that for a symmetric body, point P will be the stagnation point, which means that $U_P = 0$.

$$\lambda = -2\pi \left(\frac{1}{1.5} - \frac{1}{2.5} \right)^{-1}$$

$U_\infty = 1 \text{ m/s}$



Flow over Rankine's oval

- We know that for a symmetric body, point P will be the stagnation point, which means that $U_P = 0$.
- This gives the strength of the source and sink to simulate a Rankine's oval in the flow.
- The same approach is used to simulate all kinds of bodies in a flow (for example airfoils).

$$\lambda = -2\pi \left(\frac{1}{1.5} - \frac{1}{2.5} \right)^{-1}$$



Today, we have

- Constructed a potential flow solution by combining different elementary flows
- Defined the strength of the elements by setting a boundary condition



Rotor and Wake Aerodynamics

Lifting line model and its application to
a rotor and wake simulation



Learning objectives

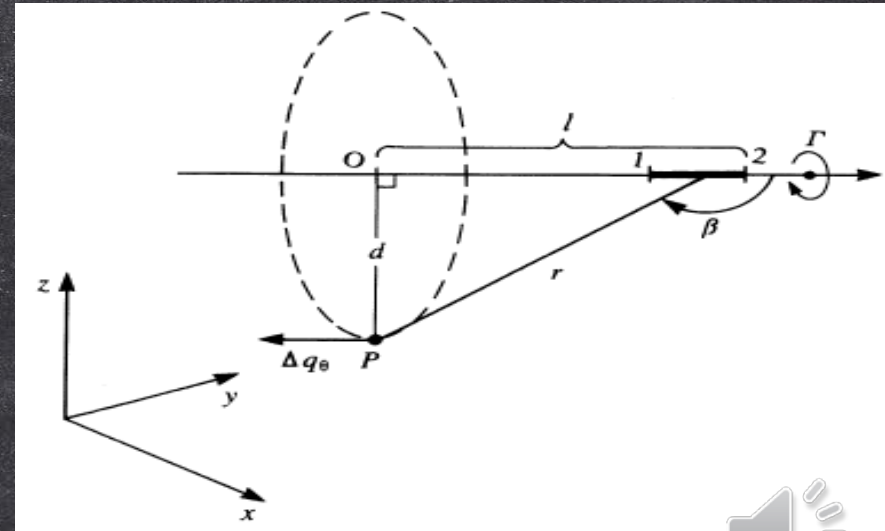
- Identify the main elements of a lifting line model
- Define the boundary conditions necessary to setup the model
- Discuss options to simplify the lifting line model



Vortex Line/Filament Methods

In line vortex or vortex filaments, vorticity is concentrated along material lines of constant circulation

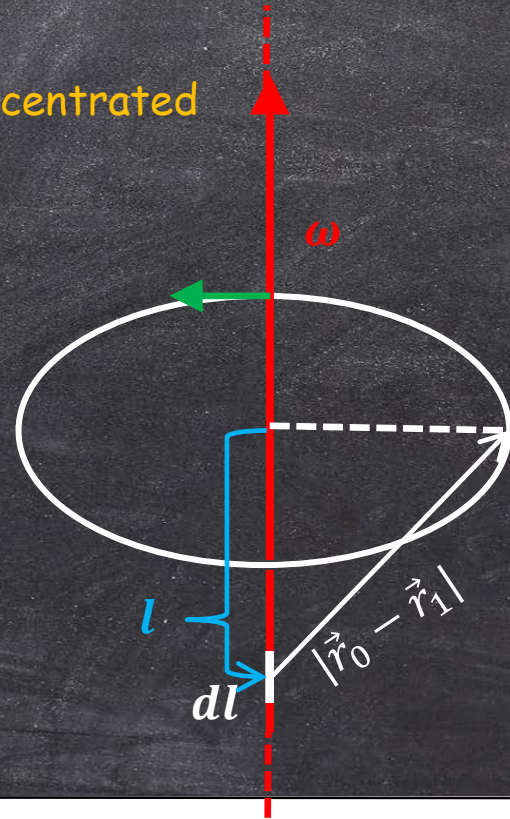
$$\vec{v}_{(\vec{x}_0)} = \Gamma \int_0^l \frac{d\vec{l}_{(\vec{x})} \times (\vec{r}_0 - \vec{r}_1)}{4\pi |\vec{r}_0 - \vec{r}_1|^3}$$



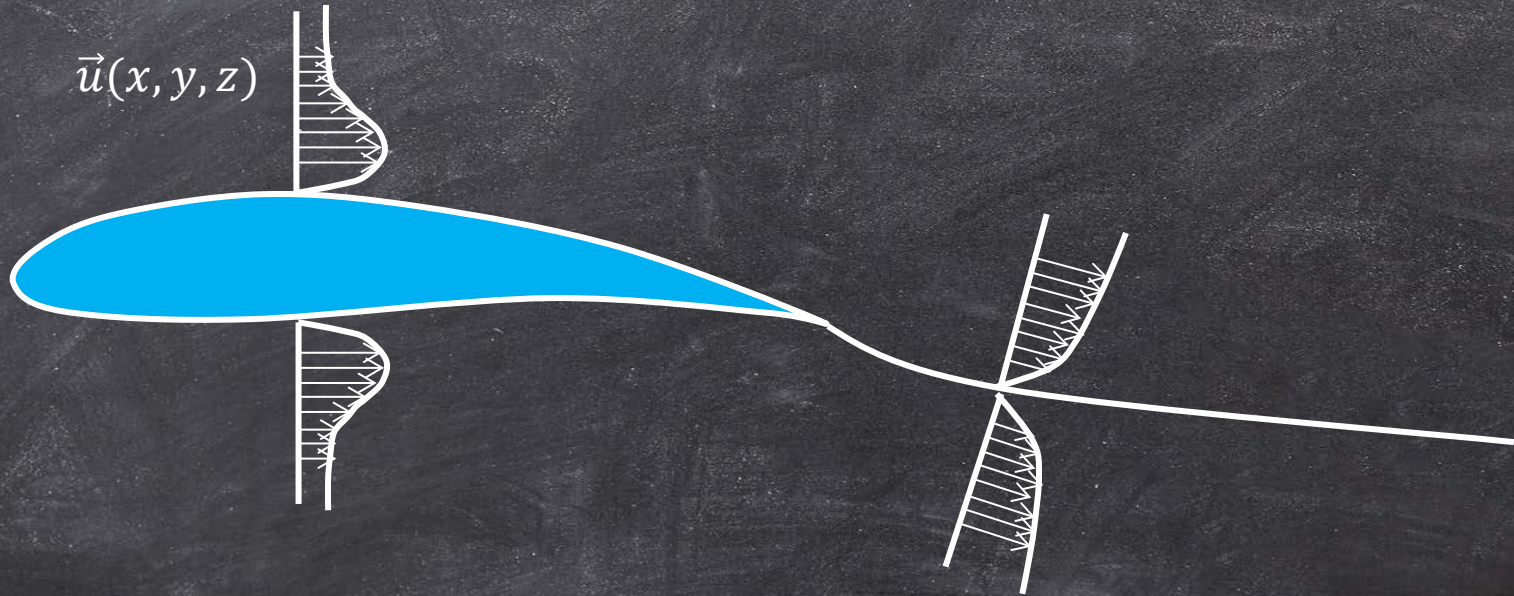
Vortex Line/Filament Methods

In line vortex or vortex filaments, vorticity is concentrated along material lines of constant circulation

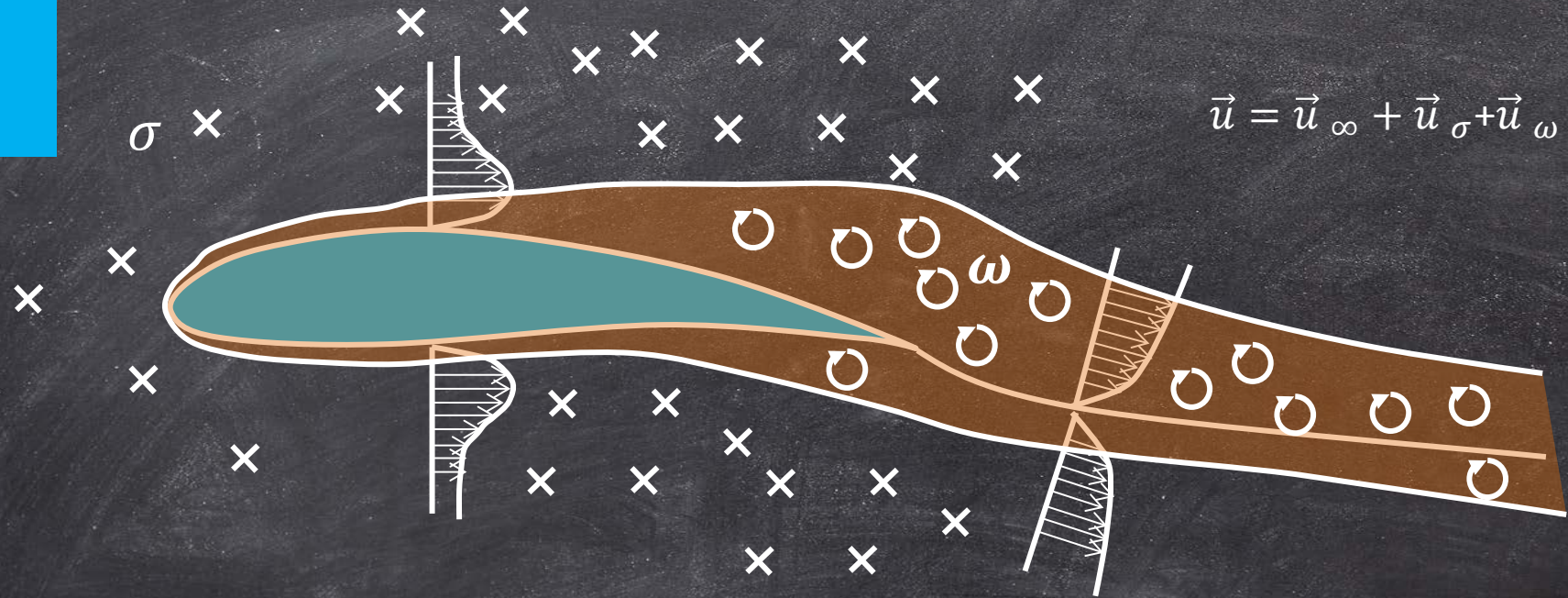
$$\vec{v}_{(\vec{x}_0)} = \Gamma \int_0^l \frac{d\vec{l}_{(\vec{x})} \times (\vec{r}_0 - \vec{r}_1)}{4\pi |\vec{r}_0 - \vec{r}_1|^3}$$



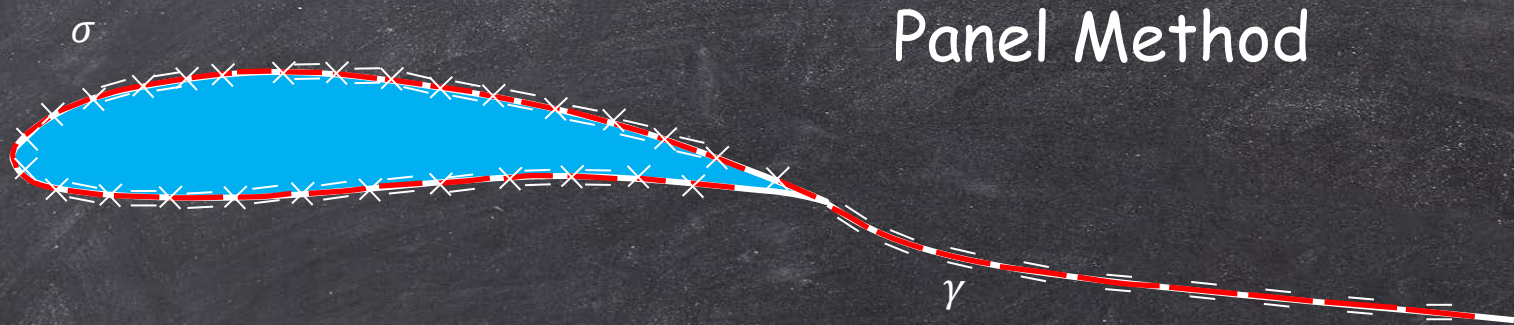
Flow Field Approximations: Panel Method, Lifting Surface, Lifting Line



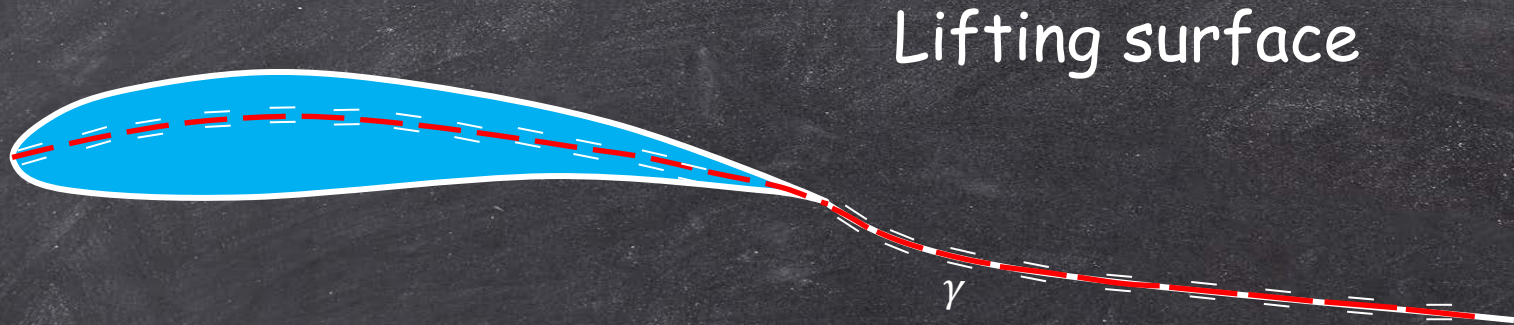
Flow Field Approximations: Panel Method, Lifting Surface, Lifting Line



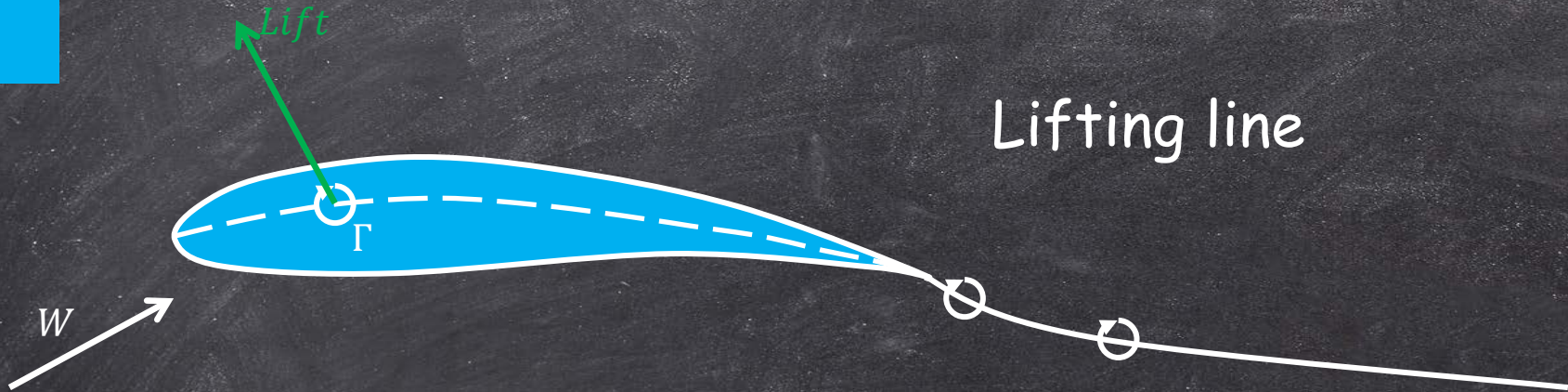
Flow Field Approximations: Panel Method, Lifting Surface, Lifting Line



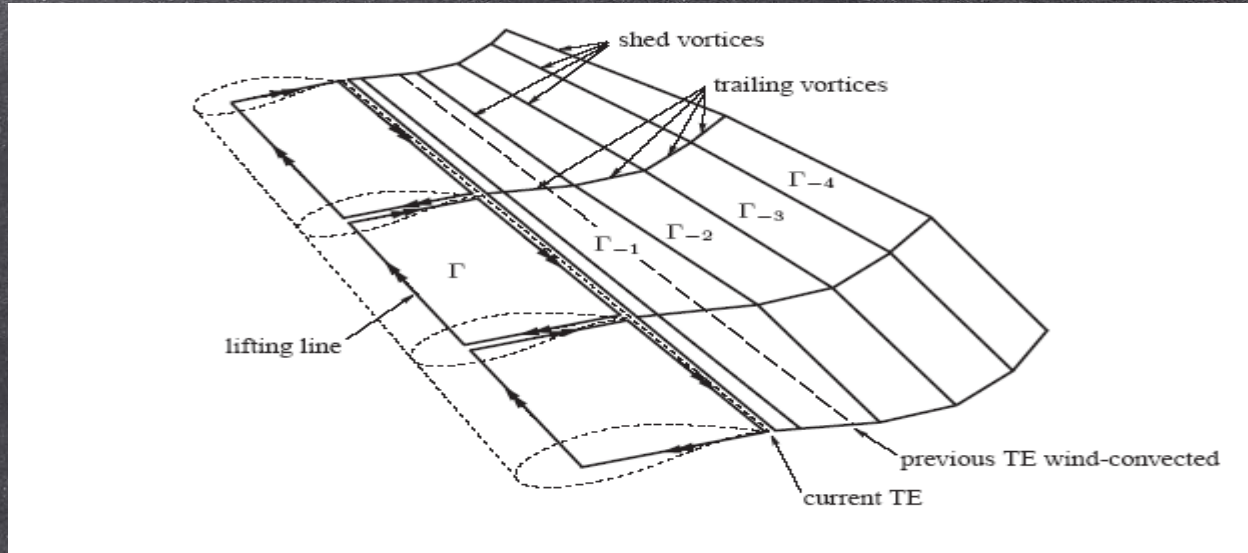
Flow Field Approximations: Panel Method, Lifting Surface, Lifting Line



Flow Field Approximations: Panel Method, Lifting Surface, Lifting Line



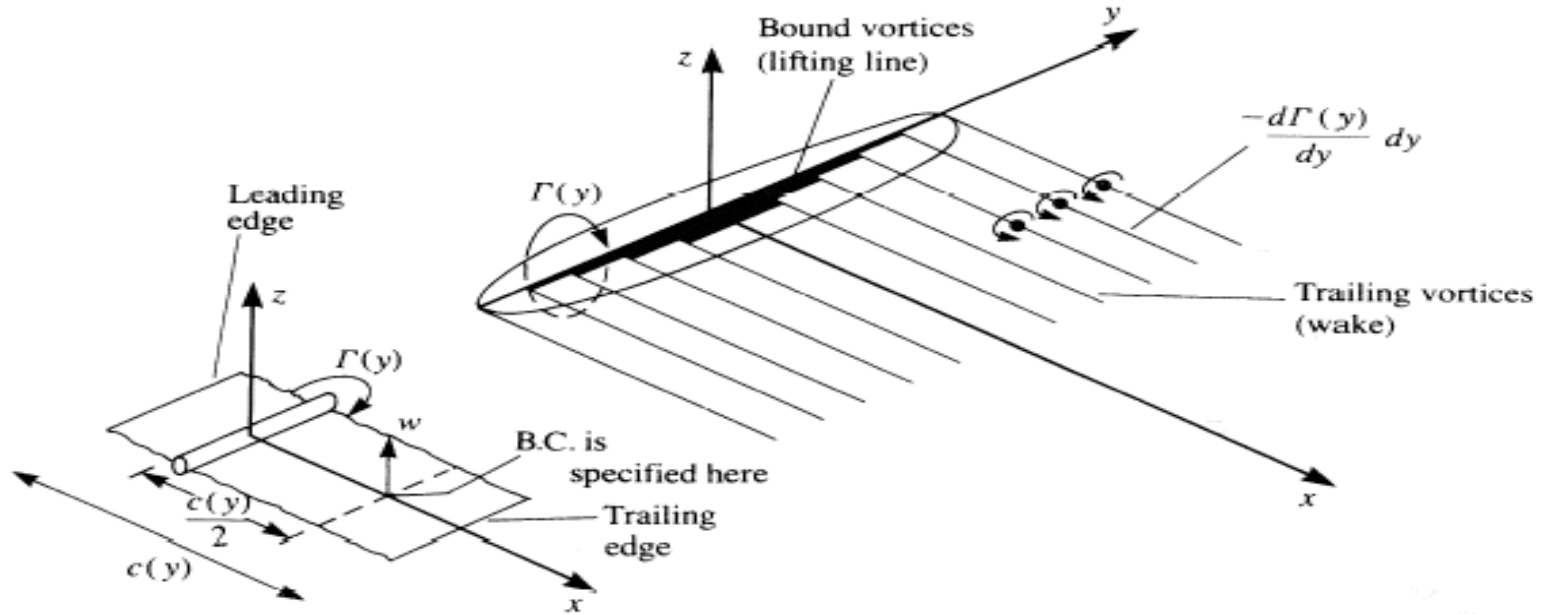
The vortex wake structure is composed of trailing and shed vortices



A. Van Garrel, "Development Of A Wind Turbine Aerodynamics Simulation Module", ECN-C--03-079, 2003



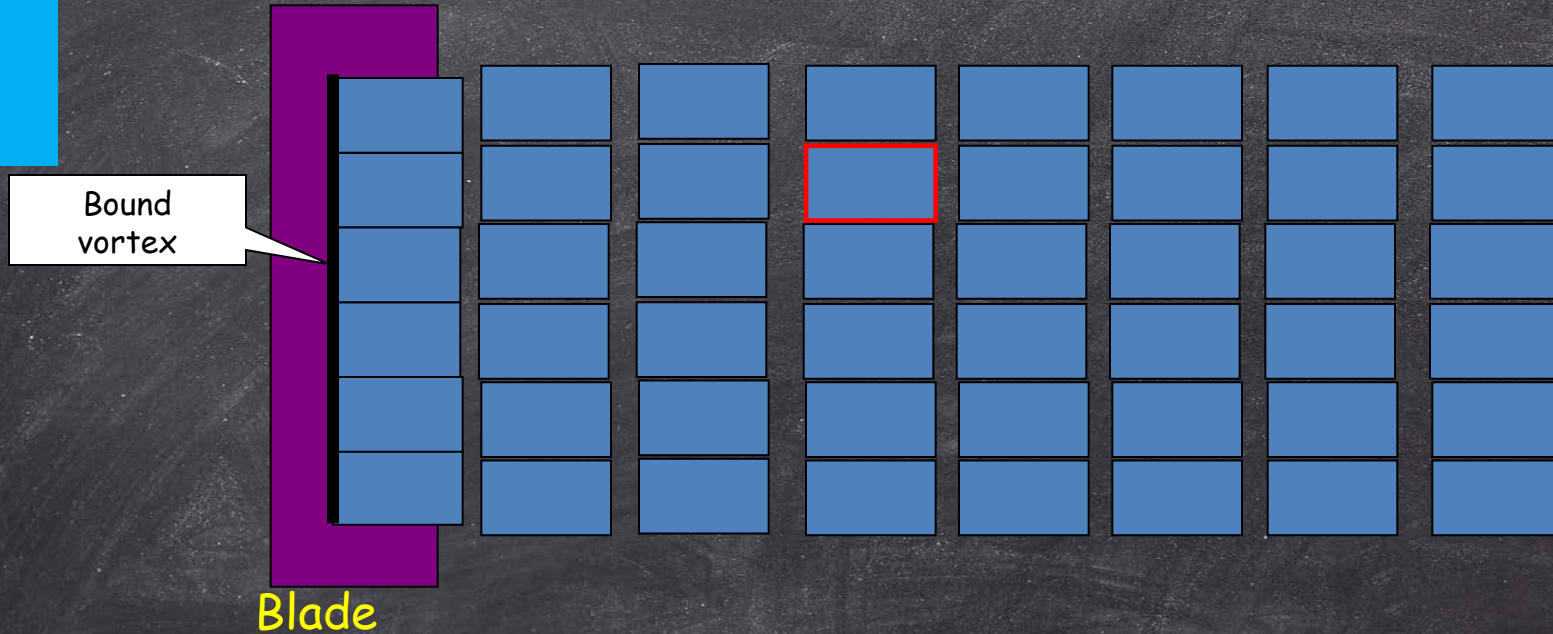
Horseshoe vortex method



Vortex wake



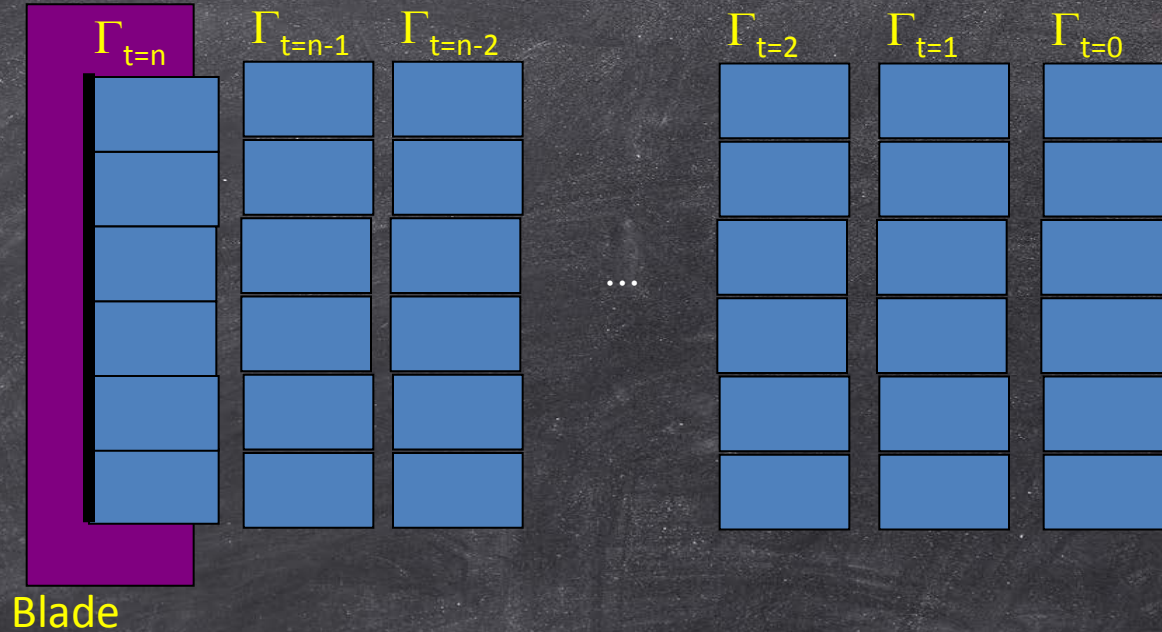
Vortex wake: ring discretization



Vortex wake: initial condition



Vortex wake: time development



Frozen vs Free Wake Vortex Model

Frozen Wake Vortex Model-

It is assumed that the geometric constraints and the unperturbed flow velocity are the only parameters affecting the wake. Wake velocity self-induction or velocities induced by the blades are neglected. No deformation of the wake occurs.

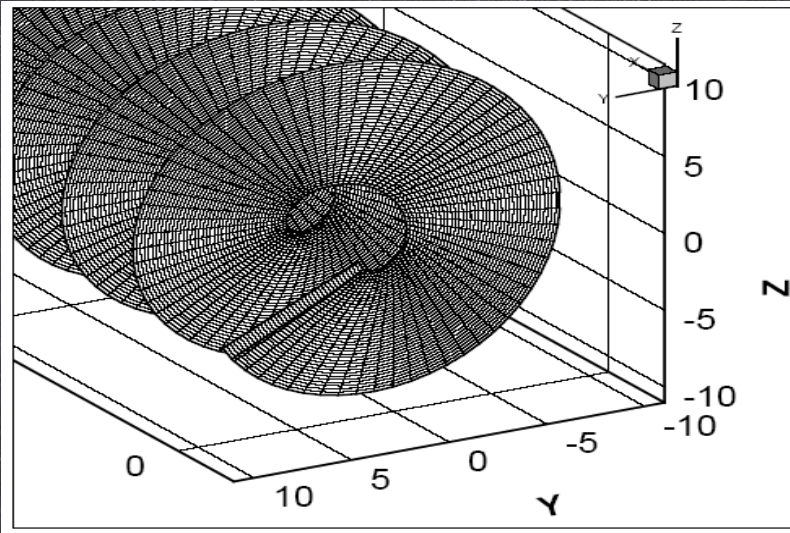
Free Wake Vortex Model-

The model takes into consideration the velocities induced on the wake by the blades' circulation and by the wake itself. Wake deformation occurs.

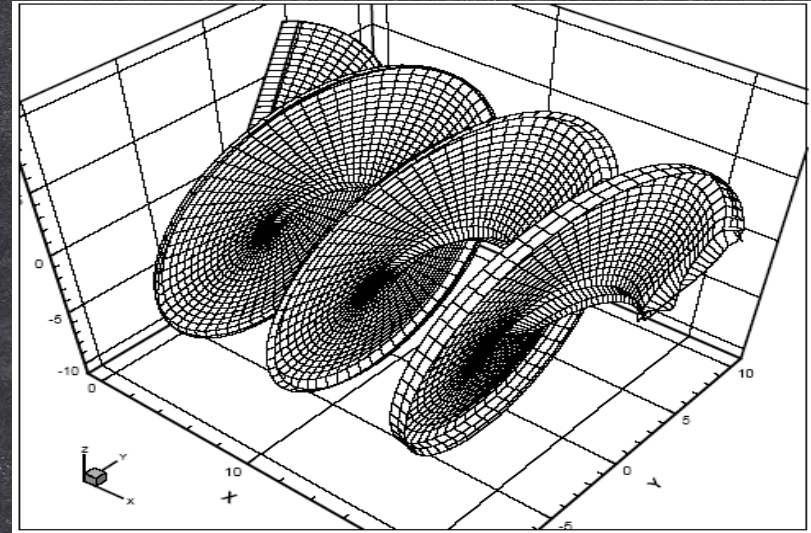


Frozen vs Free Wake Vortex Model

Frozen wake

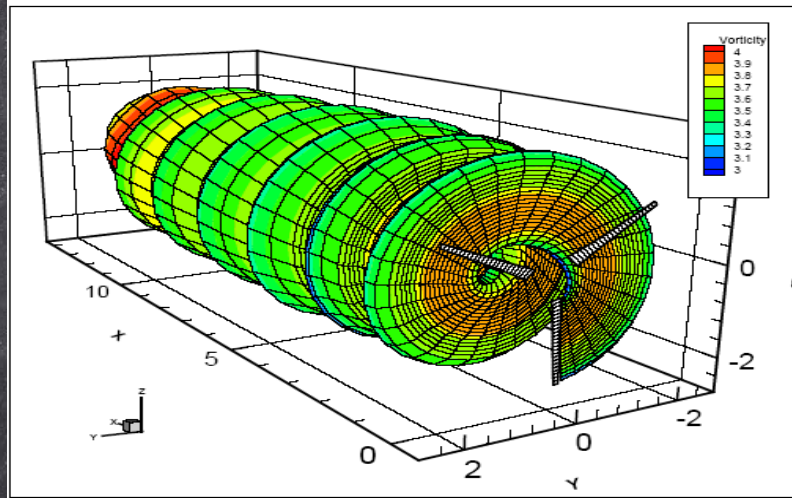


Free wake

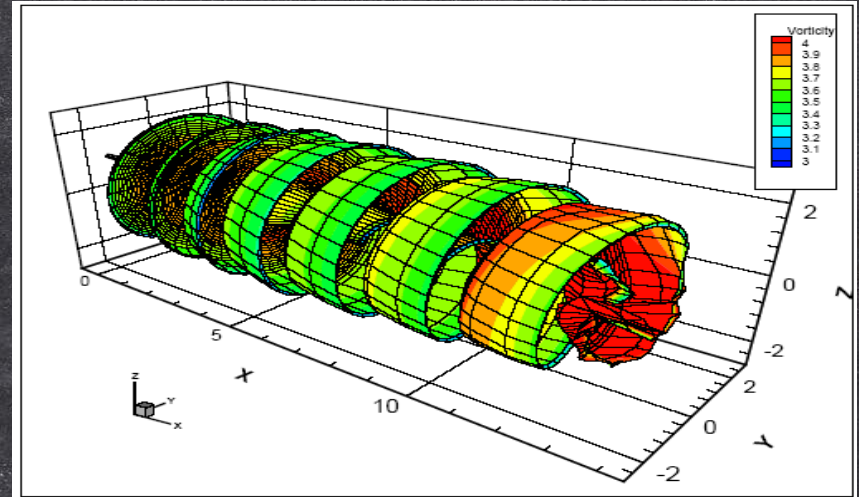


Frozen vs Free Wake Vortex Model

Front view



Back view



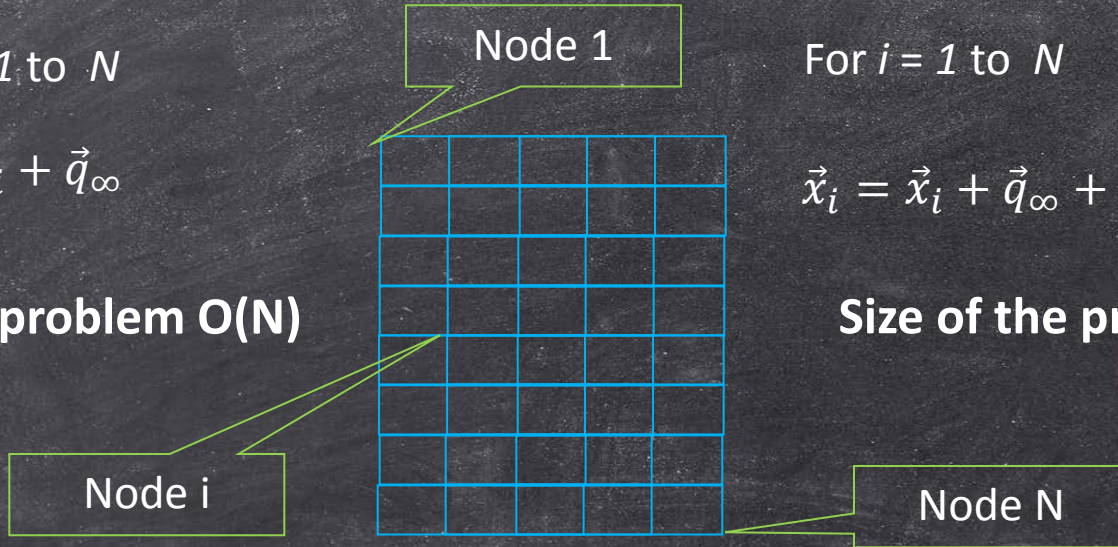
Free wake models consume much more time than frozen wake models

Frozen wake

For $i = 1$ to N

$$\vec{x}_i = \vec{x}_i + \vec{q}_\infty$$

Size of the problem $O(N)$



Free wake

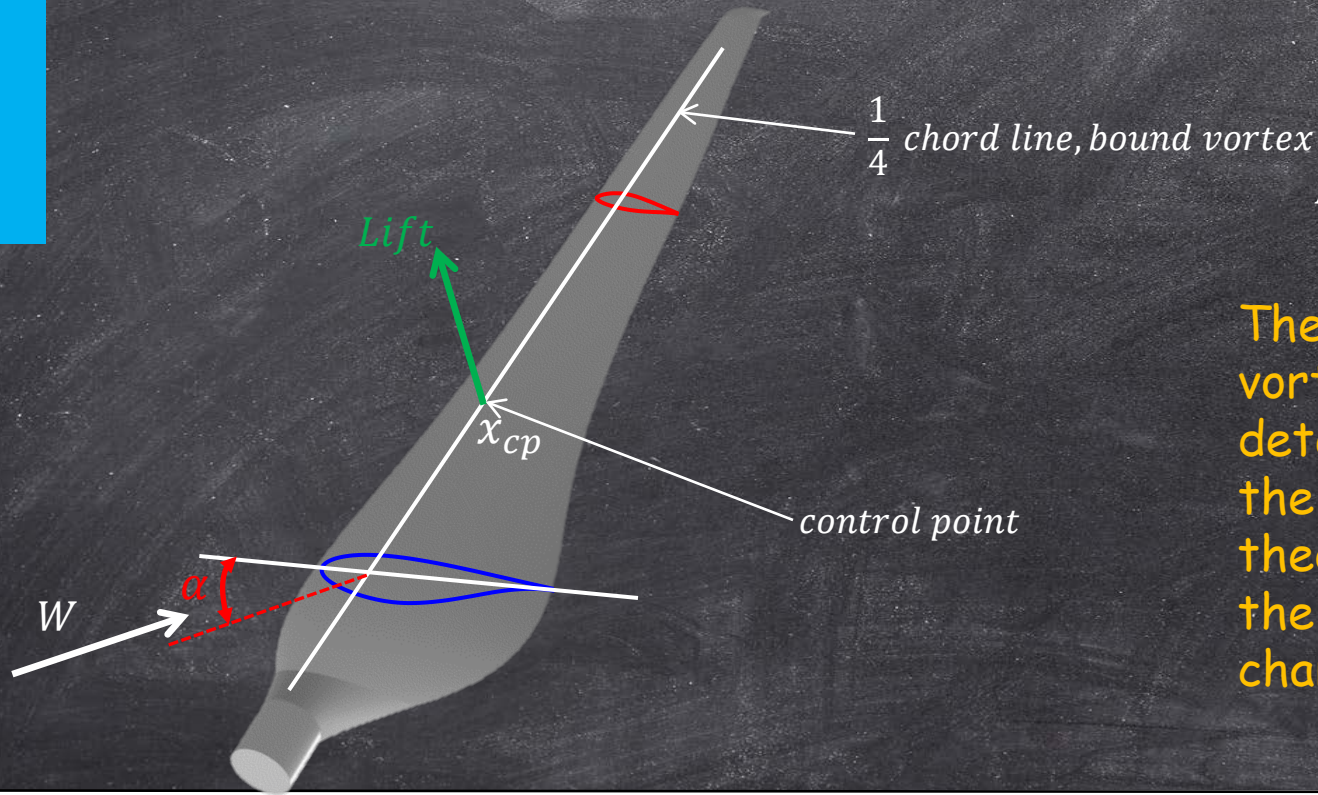
For $i = 1$ to N

$$\vec{x}_i = \vec{x}_i + \vec{q}_\infty + \sum_{j=1}^N \overrightarrow{Ind}_j(\vec{x}_i)$$

Size of the problem $O(N^2)$



The strength of the vortex line is determined locally

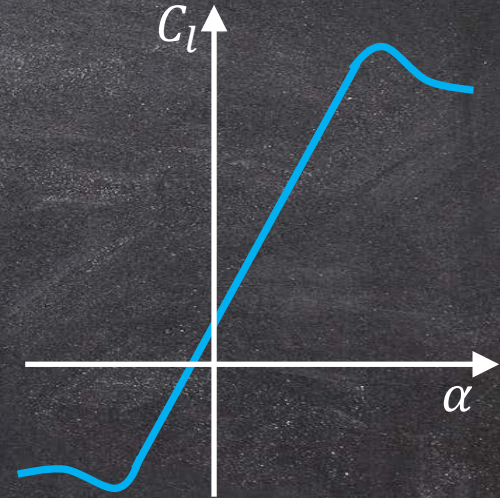
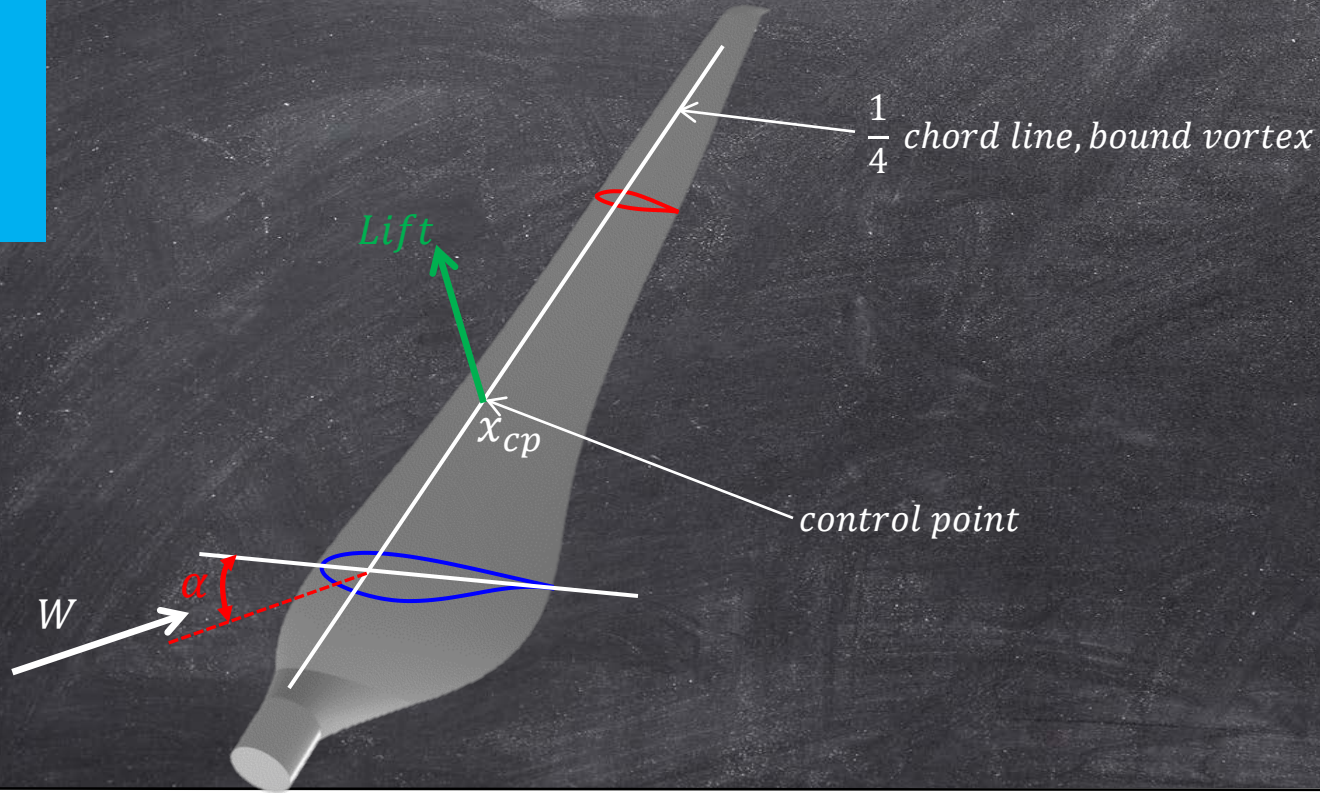


$$L = -\rho W \Gamma$$

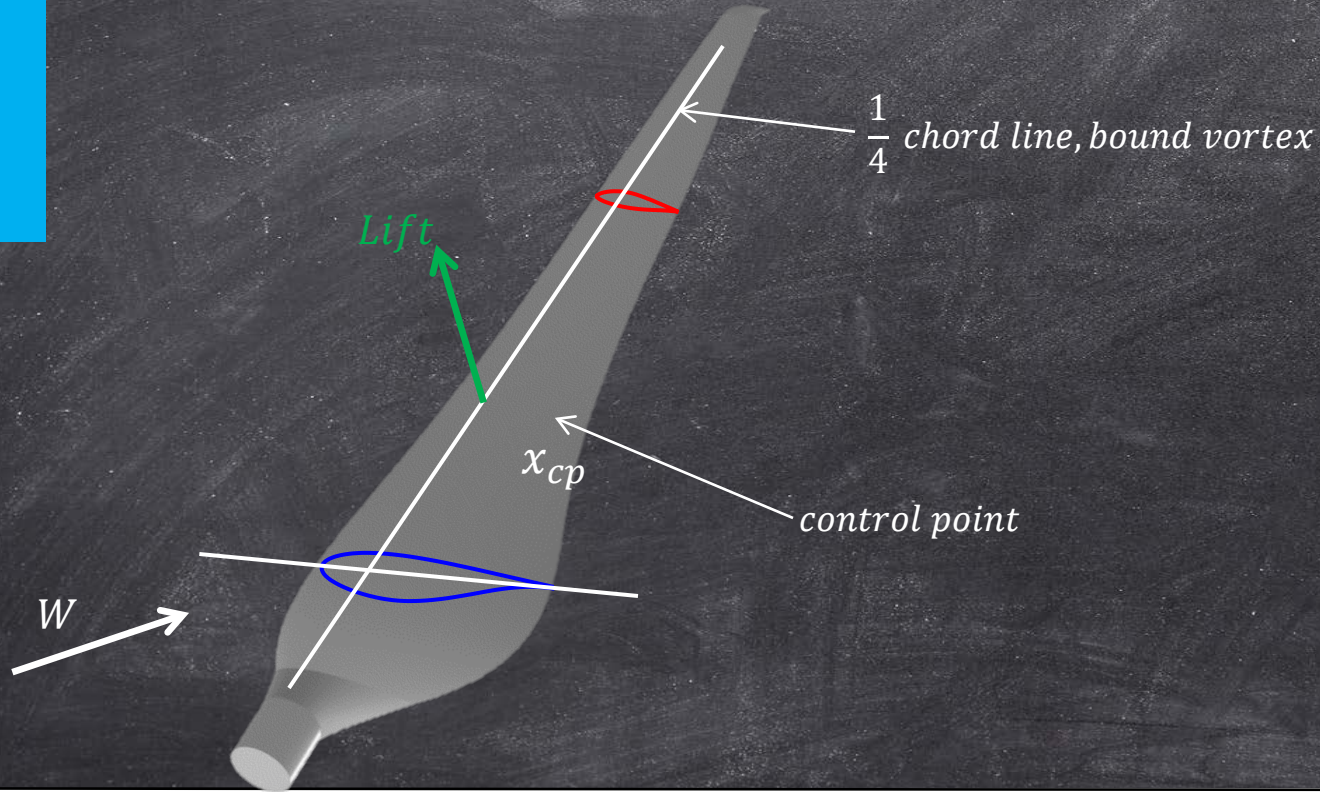
The strength of the vortex elements are determined by employing the Kutta-Joukowski theorem on the basis of the local aerofoil characteristics



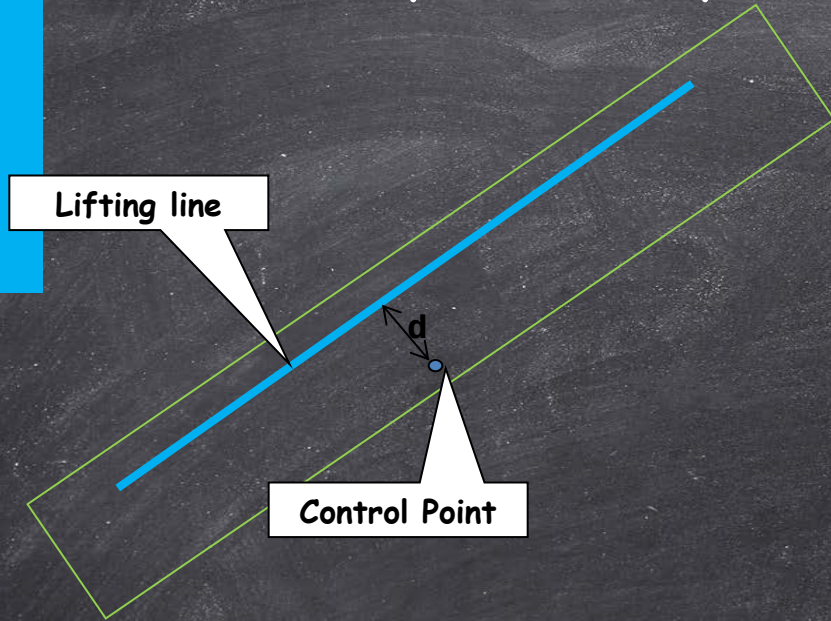
Using a angle of attack condition



Non-entry boundary condition to determine lift (I)



Non-entry boundary condition to determine lift (II)



Velocity at control point is zero
(boundary condition)

$$\vec{U}_{vor} + U_{\infty} \sin(\alpha) = 0$$

$$\vec{U}_{vor} = \frac{\Gamma}{2\pi d}$$

$$\Gamma = \frac{1}{2} U_{\infty} c C_L$$

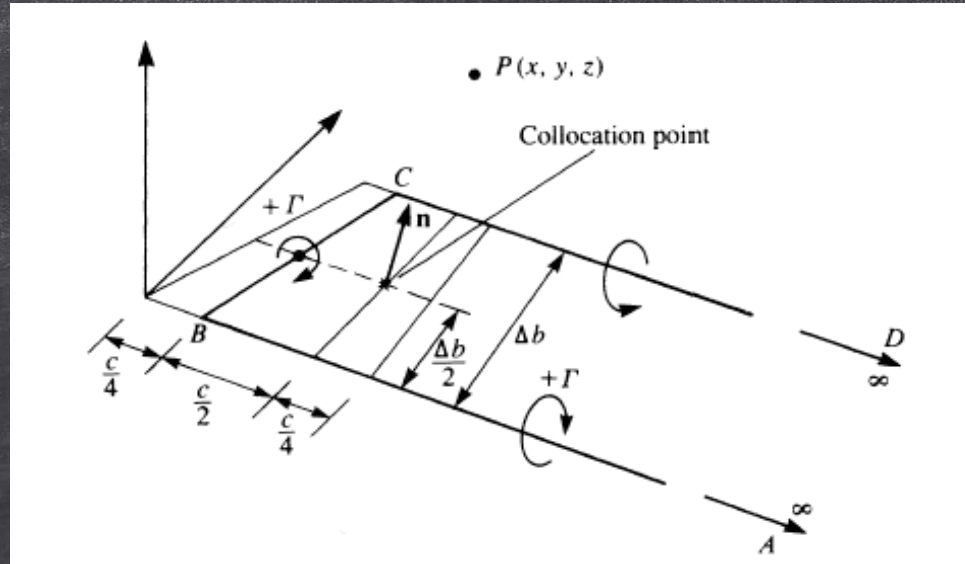
$$C_L = 2\pi \sin(\alpha)$$

Pistolessi theorem - $\frac{3}{4}$ chord

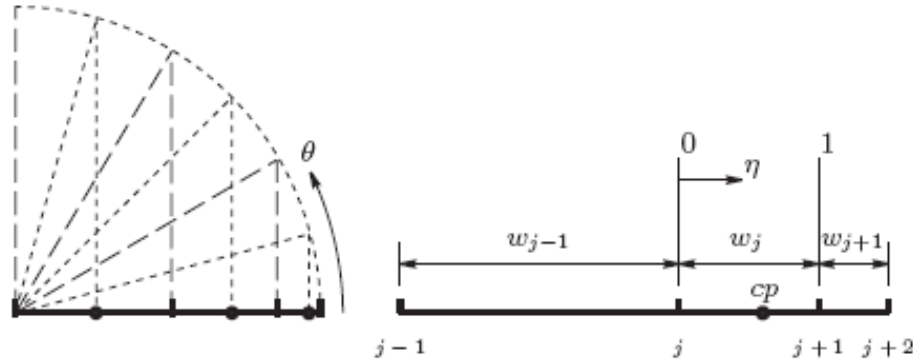
$$\Gamma = \pi U_{\infty} c \frac{\Gamma}{2\pi d U_{\infty}} \Rightarrow d = \frac{c}{2}$$



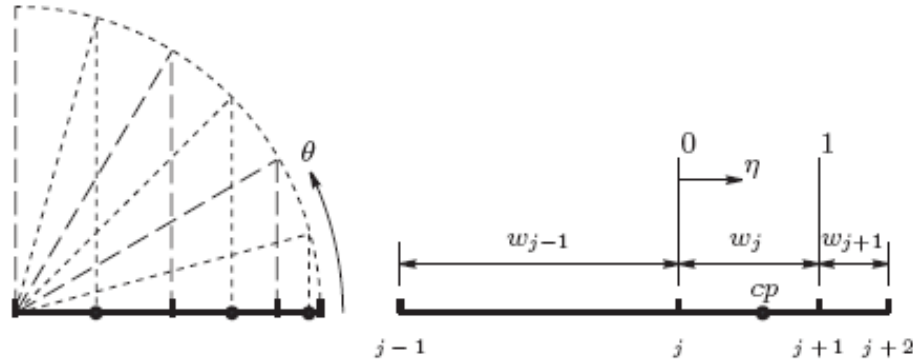
Non-entry boundary condition to determine lift (III)



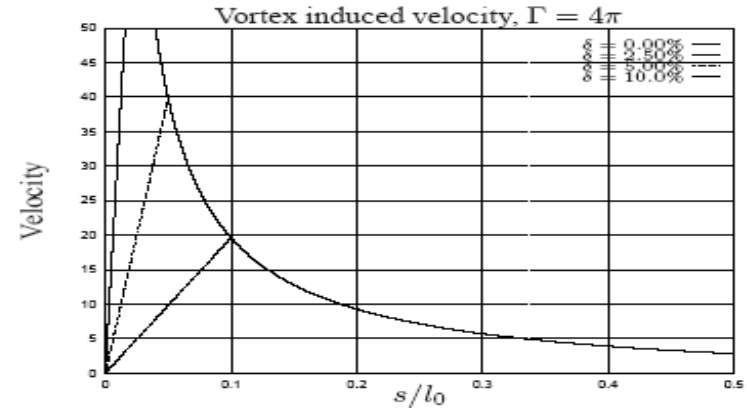
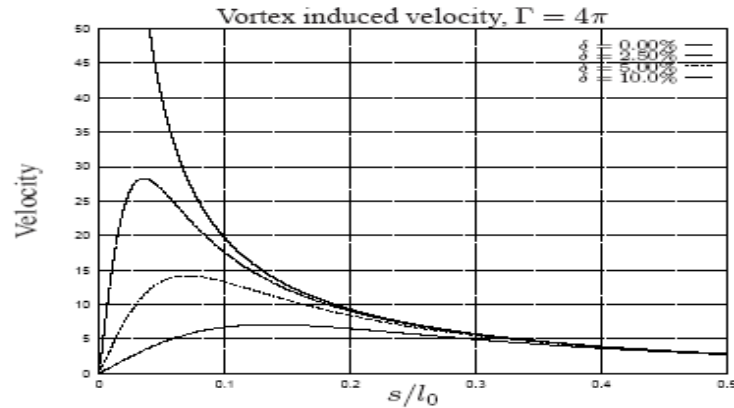
Control Points Distribution



Control Points Distribution



Modeling the Vortex Core



We have:

- Identified the main elements of a lifting line model
- Defined the boundary conditions necessary to setup the model
- Discussed options to simplify the lifting line model

